FRAMEEVIDENCE: A Benchmark for Reference Frames and Visual Evidence in Multi-Image Spatial Reasoning

Anonymous ACL submission

Abstract

A spatial answer depends on the reference frame specified in language and the visual evidence supporting it. Item accuracy alone obscures whether a model responds appropriately when either changes. We introduce FRAMEEVIDENCE, a benchmark of 320 matched triplets and 960 Chinese multiple-choice questions spanning captured and rendered scenes. Its two question families query three grounded reference frames over fixed images and pair annotated deletion-to-insufficiency with answer-preserving addition. Joint metrics complement item accuracy by requiring correct answers across complete frame groups and paired evidence variants. Across 20 configurations on 288 held-out questions, GPT three-shot reaches 50.00% item accuracy, but the best three-frame group accuracy is 5/48 (10.42%), reached by GPT two-shot. Few-shot prompting and task-specific adaptation improve item accuracy without consistently improving joint correctness. Input controls and repeated executions further distinguish evidence dependence from response stability. Together, these diagnostics reveal when individual successes extend to correctness sustained across matched changes.

1 Introduction

Spatial language describes an observed arrangement relative to a coordinate convention. An object on the left from one camera may be in front under another object's orientation. A model must also determine whether partial views jointly support that description. Spatial supervision has improved spatial question answering and regional reasoning in vision-language models (Chen et al., 2024; Cheng et al., 2024). Evaluation must therefore test both frame grounding and visual evidence use.

Existing benchmarks test perceptual and three-dimensional spatial relations (Liu et al., 2023; Fu et al., 2024; Ma et al., 2025), reference-frame ambiguity and subject-centered judgments (Zhang et al., 2025; Premsri and Kordjamshidi, 2025; Liu et al., 2026). Multi-view and video evaluations extend this coverage to evidence distributed across observations (Yang et al., 2026; Yeh et al., 2026; Yang et al., 2025). Yet item accuracy does not show whether correctness persists across related inputs.

A correct direction may reflect the specified frame or a preferred answer that happens to match it. Distinguishing these behaviors requires observing responses to changed frames or evidence, but scenes, entities and wording often vary together. Matched evaluations instead compare related inputs (Thrush et al., 2022; Wang et al., 2025; Kim et al., 2026). The essential distinction is between changes that *should alter* the answer and those that *should preserve* it.

FrameEvidence expresses this distinction through two protocols in Figure 1. The first fixes ordered images and the target–relatum pair while querying camera, object-anchor and scene-axis frames. The second starts from an image set annotated as subset-minimal and sufficient. Deleting a required view should yield insufficient information, while adding a difficult but non-necessary view should preserve the spatial answer. This pairs necessity and invariance checks within one instance, extending the evidence dependence examined by rationale, unanswerability and view-selection diagnostics (DeYoung et al., 2020; Moratelli et al., 2026; Guo et al., 2026; Wang et al., 2026b).

Joint correctness requires correct answers across matched variants. Examining answer transitions separates initial errors from failures to adjust to a changed reference frame or image set.

The release contains 320 triplets and 960 questions, with 96 complete triplets reserved for testing. The main evaluation covers ten open-weight checkpoints, two adapted counterparts and two gateway-served aliases under zero- to three-shot prompting. The strongest item-level configuration answers half the test correctly, yet only three of its 48 reference-

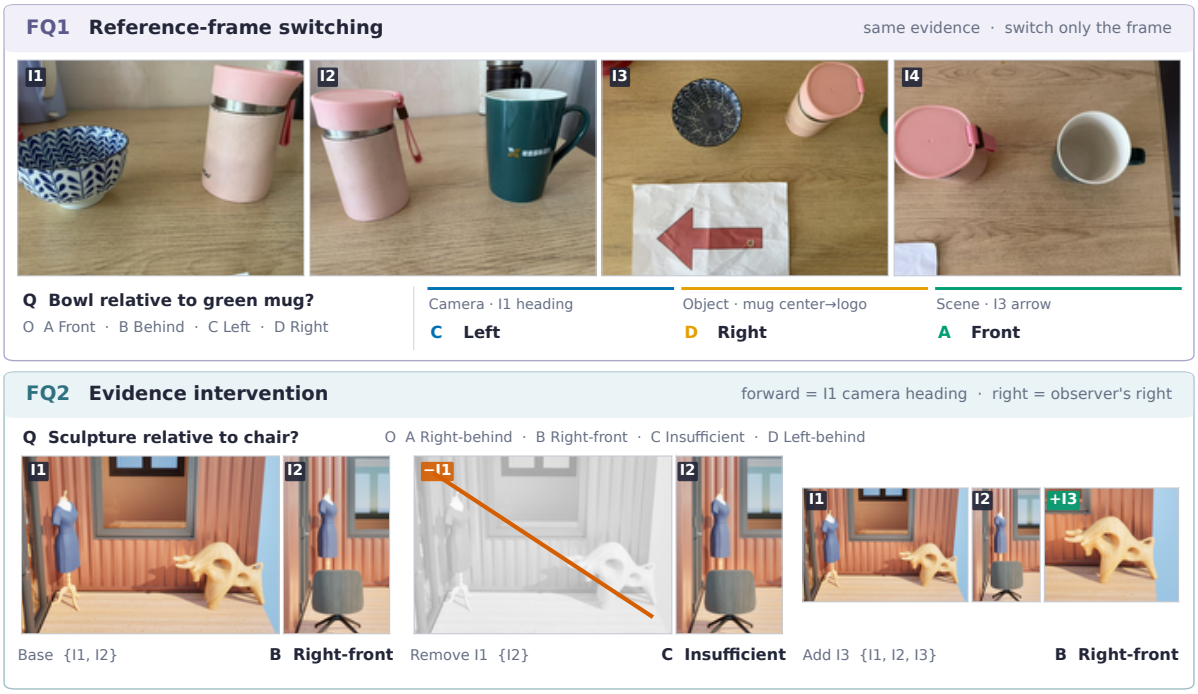


Figure 1: Matched protocols. FQ1 changes the queried frame while fixing the images. FQ2 deletes an annotated required view or adds a hard non-MSS view. The crossed-out image is withheld, and I1–I4 mark original positions for display. After I1 is deleted, "the first image" refers to I2.

frame groups are entirely correct. Input controls, adaptation comparisons and option rotations explain why similar item scores can conceal different responses to matched changes.

FrameEvidence distinguishes individual gains from matched success, with dataset analysis tracing the condition cues and image references involved.

2 Related Work

Table 1 compares the evaluation protocols below.

Reference frames and frame-conditioned reasoning. Spatial and matched-contrast benchmarks probe relational judgments (Liu et al., 2023; Fu et al., 2024; Thrush et al., 2022). COMFORT, FoREST, and ViewSpatial-Bench examine reference-frame ambiguity (Zhang et al., 2025; Premsri and Kordjamshidi, 2025; Li et al., 2026). SCOPE, All-Angles, and CityCube address cross-view and scale variation (Kim et al., 2026; Yeh et al., 2026; Xu et al., 2026), while MirrorQA targets subject-centered left–right judgments (Liu et al., 2026). COMFORT compares explicit frames on matched synthetic scenes and measures probabilistic consistency. Our frame protocol instead scores joint correctness across three multi-view frame prompts, distinguishing a correct switch from an answer change alone. The evidence protocol similarly tests responses to changed views.

Multi-image reasoning and evidence interventions. MIRB, ReMI, MMIU, and MuirBench assess multi-image reasoning, while MIMIC probes controlled failure cases (Zhao et al., 2024; Kazemi et al., 2024; Meng et al., 2025; Wang et al., 2025; Das et al., 2026). MMSI-Bench, CrossView Suite, MultiView-Bench, and cross-view relation learning focus on spatial integration (Yang et al., 2026; Wang et al., 2026a; Zhang et al., 2026; Jeong et al., 2026). Evidence-centered diagnostics examine input dependence through rationale sufficiency, paired unanswerability, view selection, evidence sensitivity, and multi-view necessity (DeYoung et al., 2020; Moratelli et al., 2026; Wang et al., 2025; Guo et al., 2026; Wang et al., 2026b; Bhat and Yamasaki, 2026; Yu et al., 2026). SSMNBench adds and removes views of minimal necessary sets to measure view-wise accuracy and distraction decay. These measures describe supporting views and average sensitivity to evidence changes. Joint correctness asks whether success is sustained on the same instance rather than distributed across different instances. Our paired scores therefore require correct answers before and after each intervention. Comparing models on shared, correctly answered

Table 1: Comparison of evaluation protocols most relevant to FrameEvidence.

Benchmark	Src.	MI	Match	FoR S	Min Ev	Del. $\to \perp$	Hard+	Joint	Scale
COMFORT (Zhang et al., 2025)	Syn.	–	**Y**	**Y**	–	–	–	*P*	58.3K English cases
MuirBench (Wang et al., 2025)	Mix.	**Y**	**Y**	–	–	–	–	*P*	2,600 instances
MMSI-Bench (Yang et al., 2026)	Real	**Y**	–	–	*P*	–	–	–	1,000 QA
All-Angles (Yeh et al., 2026)	Real	**Y**	**Y**	*P*	–	–	–	**Y**	2,132 QA
SCOPE (Kim et al., 2026)	Mix.	**Y**	**Y**	*P*	–	–	–	*P*	20,144 QA
MultiView-Bench (Zhang et al., 2026)	Syn.	**Y**	*P*	–	–	–	–	–	500 main (+2K var.)
SSMNBench (Guo et al., 2026)	Real	**Y**	**Y**	–	**Y**	–	*P*	–	3,300 QA
View-Level Evidence (Wang et al., 2026b)	Real	**Y**	*P*	–	*P*	–	–	*P*	122 QA
FrameEvidence	**Mix.**	**Y**	**Y**	**Y**	**Y**	**Y**	**Y**	**Y**	**960 QA (320 triplets)**

Legend. **Y**/*P*/– denote explicit, partial, and unspecified coverage in the reported protocols. MI, Match, FoR-S, and Min-Ev abbreviate multi-image, item matching, frame switching, and minimal evidence. Del.$\to \perp$ denotes deletion to insufficiency, Hard+ denotes truth-preserving hard non-MSS addition, and Joint denotes correctness across matched variants. Partial Joint coverage includes response consistency or joint view–answer prediction. Scales use source units.

minimal sufficient sets (MSSs) then separates intervention errors from changes in initial coverage.

3 Problem Formulation

Each item is $x = (\mathbf{I}, q, \mathcal{O})$, where $\mathbf{I} = (I_1, \ldots, I_m)$ contains one to five images, q is a Chinese question, and $\mathcal{O} = (o_A, o_B, o_C, o_D)$ is a four-option answer set. A model returns $\hat{y} = h_\theta(\mathbf{I}, q, \mathcal{O}) \in \{A, B, C, D\}$. Spatial answers use axial (front, back, left, right) or quadrant directions (front-left, front-right, back-left, back-right). FQ2 includes insufficient information. Metric reconstruction, vertical relations and motion are excluded.

Reference-frame intervention (FQ1). Each group queries the same target–relatum relation in three frames $\mathcal{F} = \{\mathrm{cam}, \mathrm{obj}, \mathrm{scene}\}$. The camera frame uses a specified camera heading, the object frame a named object's orientation, and the scene frame an environmental axis. The question specifies the anchor, which need not be the relatum, and its forward or backward convention. Ordered images, entity identities, relation ontology and option meanings remain fixed. Only the frame instruction and its implied answer change. At least two gold answers differ, so repeating one answer cannot solve the group.

Evidence intervention (FQ2). The intervention is designed to vary evidence while fixing the target–relatum relation and reference frame. The base set M_g is annotated as a minimal sufficient set (MSS), supporting spatial answer y_g when no proper subset does. Deletion removes an annotated required image r_g. Addition restores the MSS and adds a hard, truth-preserving image $d_g \notin M_g$. The annotated responses to M_g, $M_g \setminus \{r_g\}$ and $M_g \cup \{d_g\}$ are respectively y_g, $\perp$ and y_g, where $\perp$ means insufficient information. Subset-minimality does not imply minimum size. Section 5 examines the resulting evidence and reference changes.

Group-level evaluation. Let $c_g^v = \mathbb{1}[\hat{y}_g^v = y_g^v]$ denote correctness for variant v, and $n_k = |\mathcal{G}_k|$ the number of groups in family k. FQ1 Group Exact Match scores the three frame answers jointly,

$$\text{FQ1-GEM} = n_1^{-1} \sum_{g \in \mathcal{G}_1} \prod_{f \in \mathcal{F}} c_g^f. \qquad (1)$$

For FQ2, Necessity Transition Success is $\text{NTS} = n_2^{-1} \sum_g c_g^M c_g^-$, and Distractor Invariance Success is $\text{DIS} = n_2^{-1} \sum_g c_g^M c_g^+$, with both sums over $\mathcal{G}_2$. NTS scores agreement with both the MSS and deletion labels, while DIS requires the spatial answer to remain correct after addition.

4 FrameEvidence Construction

4.1 Scene Acquisition and Group Assembly

FrameEvidence combines 126 captured-image and 194 rendered-image groups of static layouts spanning indoor tabletop, indoor large-space and outdoor public-space scenes. An object inventory identifies the target, relatum and any bridge objects connecting views. Construction criteria require unambiguous identity matches, unchanged layouts, grounded frame anchors and evidence that must be combined across views.

FQ1 reuses two to four ordered images across its three questions. FQ2 starts from two to four MSS images, then changes the set size by exactly one per intervention. Each family allocates 53/54/53 base sets of two, three and four images. Hard additions introduce competing context while preserving the relation. The checklist covers answer preservation and the necessity of each MSS image.

Table 2: Dataset composition by split. All entries count questions except the Groups row.

Category	Train	Dev	Test	Total
Groups	192	32	96	320
Questions	576	96	288	960
FQ1	288	48	144	480
FQ2	288	48	144	480
Tabletop	195	27	102	324
Indoor large	192	33	93	318
Outdoor public	189	36	93	318
Axial	291	45	144	480
Quadrant	285	51	144	480

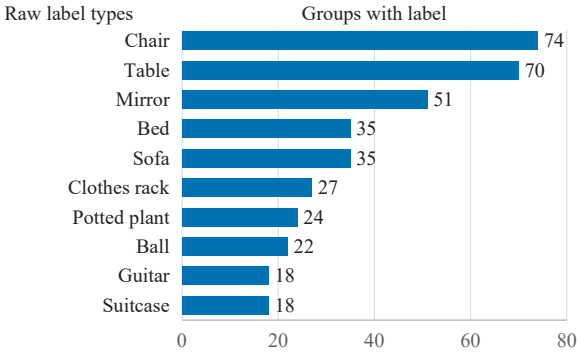


Figure 2: Object-label counts per group. The other 116 labels contribute 586 of the 960 label occurrences.

4.2 Annotation and Quality Control

Controlled Chinese questions specify the queried entities and reference frame. A constrained placement step maps semantic answers to A–D, sharing option order within groups. Each letter occurs 240 times overall and 40 per condition.

The production ledger records approval for each group by a separate reviewer account with answers visible. Its checklist covers scenes, bindings, directions, MSS necessity and answer preservation. Automated schema and consistency checks complement this review, documented in Appendix A.

4.3 Split and Validation Scope

Table 2 allocates 192/32/96 complete groups to training/development/test. Matched variants and identical images stay together, preventing training and test questions from sharing a triplet. Different views may nevertheless share a scene across splits. The split therefore evaluates held-out groups without enforcing scene isolation.

Table 3 reports one participant's sampled blind-answer summary. Appendix A describes this reference and the exact random baseline.

5 FrameEvidence Analysis

Composition and balance. Table 2 shows nearly equal scene strata and balanced relation ontologies, with 160 questions per condition. The ten most frequent labels in Figure 2 cover 374/960 group-level occurrences. The other 116 retain source-specific color and naming variants, as detailed in Appendix B. Test A–D counts are 75, 70, 69 and 74, motivating macro-averaging.

Evidence structure and non-visual controls. Figure 3a shows that extreme image counts identify some FQ2 conditions. On 144 held-out FQ2 items, a train-fitted count-plus-schema rule achieves 37.50% accuracy, compared with 33.33% from always selecting insufficient information. Yet its NTS and DIS are 6.25% and 2.08%, revealing little success across paired variants. A character TF-IDF classifier using question and option text reaches 23.96% accuracy across all 288 test items. Identical text across FQ2 variants forces this deterministic classifier to repeat its answer, making NTS zero by construction. These controls show why item accuracy alone cannot establish success across matched evidence changes. Appendix F.3 details the text baseline.

Scene overlap and sensitivity. The release has 443 distinct images across 1,120 paths. Exact images stay within one split, although views and renderings can share scenes across splits. An inferred scene/template ledger assigns 81 test groups to clusters shared with training and 15 to four clusters absent from both train and dev. On this ledger-nonoverlap subset, Qwen's primary LoRA lowers DIS from 4/8 to 2/8, retaining the aggregate direction. InternVL's NTS instead changes in opposite directions across strata, which cancel in the aggregate. With 13 synthetic and two captured-image groups, the subset also differs in scene composition. Appendix F.1 reports all configurations.

Ordinal references after deletion. Of 48 FQ2 test groups, 44 name an image by ordinal. Deletion removes the original referent in 36 groups and shifts it in six. Under the current-order, unlabeled presentation, 38 ordinals resolve to another retained image and four fall out of range. Evidence removal therefore also changes the question's anchor in these cases. NTS measures released-label agreement under this combined change. Appendix E traces these dependencies group by group.

(a) Input-image counts

FQ2 release: 160 groups / 480 items

MSS (groups)	MSS	Deletion	Addition
2 images (53)	2	1	3
3 images (54)	3	2	4
4 images (53)	4	3	5

One matched MSS / deletion / addition triplet per group.

(b) Held-out metadata controls

FQ2 test: 144 items / 48 groups; train-fitted

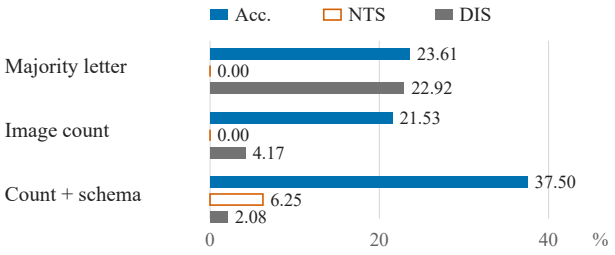


Figure 3: FQ2 evidence and held-out metadata controls. Image counts cover the release in (a). Rules in (b) fit only training groups. Test accuracy uses 144 items, while NTS and DIS use 48 groups. Schema denotes the visible insufficient-information option's letter.

Reference cues also affect FQ2 frame strata. Camera headings occur in 30/48 MSS prompts with non-camera tags, so the strata describe annotation groupings rather than distinct frame conditions.

6 Experiments

6.1 Evaluation Setup

Table 3 compares ten open-weight checkpoints, two LoRA variants and eight gateway configurations on 288 questions. GPT-5.6 and Claude Fable 5.1 are the requested gateway aliases.

Main and input-control evaluations share the primary checkpoints. Table 5 separately compares six adapters with matched bases in ordinary and rotation runs, alongside GPT zero-shot.

Queries preserve image order and request a single letter. Open-weight and Claude runs use temperature zero and 32 output tokens. GPT instead leaves temperature unset and allows 2,048 output tokens with reasoning effort set to low.

Following MirrorQA (Liu et al., 2026), we evaluate zero- to three-shot prompting. Demonstrations use nested prefixes of one training group per family, selected with seed 20260722, excluding test groups and exact test images.

We report accuracy, macro-precision, recall and F1 over A–D on all 288 questions, with joint metrics over each family's 48 groups. Ten LLaVA-OneVision responses are invalid and count as errors. All other main responses are valid.

6.2 Item Accuracy and Matched Success

The best item-level setting need not maximize matched-group performance. GPT three-shot reaches **50.00%** accuracy, while two-shot leads FQ1-GEM and NTS at 10.42% and **22.92%**. In Table 3, MiniCPM-V-4.5 matches GPT three-shot's **47.92% DIS** despite lower accuracy of 32.99%.

This divergence depends on which answers are correct together. Independent cross-group rearrangement preserves each variant's accuracy and gives their product as a reference. NTS falls below this reference in 19/20 configurations, whereas DIS exceeds it in all 20. The image-free training-majority control reproduces both directions. Addition preserves the gold answer, while deletion changes it to insufficiency. Repeating the MSS answer can therefore succeed on addition yet miss the required deletion transition. Appendix D.4 details how shared answers contribute to this contrast.

6.3 Input Controls and Adaptation

Text-only removes all images and single-image keeps the first for twelve open-weight configurations and GPT zero-shot. Questions, options and labels stay fixed in Table 13.

Visual input most consistently improves addition-pair success. Images raise DIS for 11/12 open-weight configurations but accuracy for only six in Table 4. Qwen2.5-VL-7B gains **43.8 DIS points**, with a 95% interval of [29.2, 59.5], despite lower accuracy. GPT gains **8.7 accuracy points** over text-only, which always answers insufficiency in FQ2. Its DIS rises from 8.33% with the first image to 35.42% with all images. Thus visual evidence can improve matched success without improving overall item accuracy.

Adaptation, however, does not preserve this visual benefit. Across three seeds in Table 5a, Qwen reaches **36.00±1.40%** accuracy, yet every seed solves 6/48 addition pairs versus 22/48 for its control. The separate primary checkpoints of both families also show DIS declines, with intervals

Table 3: All-images model results (%) on 288 questions, with 48 groups per family for joint metrics.

		Classification				Matched success		
Model	Setting	P	R	F1	Acc.	GEM	NTS	DIS
Open-weight baselines								
Qwen2.5-VL-7B	Base	31.06	29.43	25.99	28.82	0.00	2.08	43.75
InternVL3-8B	Base	29.72	29.84	28.65	29.51	2.08	8.33	20.83
Qwen3-VL-8B	Base	34.60	34.30	33.83	34.03	0.00	14.58	39.58
LLaVA-OneVision-7B	Base	30.56	29.33	29.04	29.17	0.00	0.00	37.50
MiniCPM-V 4.5	Base	33.48	33.33	32.27	32.99	0.00	4.17	**47.92**
Mistral Small 3.2 24B	Base	26.99	25.96	24.82	25.69	0.00	2.08	8.33
Qwen2.5-VL-32B	Base	33.29	33.56	32.15	33.33	0.00	6.25	35.42
InternVL3.5-14B	Base	28.14	28.35	27.86	28.12	0.00	6.25	25.00
Phi-4-Multimodal	Base	26.07	26.61	25.54	26.39	0.00	0.00	35.42
Kimi-VL-A3B	Base	36.74	29.48	26.73	29.17	0.00	0.00	39.58
Task-adapted counterparts								
Qwen2.5-VL-7B	LoRA	35.45	35.53	34.81	35.42	0.00	12.50	12.50
InternVL3-8B	LoRA	34.35	32.91	30.99	32.64	0.00	8.33	8.33
Gateway-served models								
GPT-5.6	0-shot	40.57	40.72	40.43	40.62	4.17	14.58	35.42
	1-shot	44.01	44.26	44.05	44.10	6.25	16.67	41.67
	2-shot	48.53	48.29	48.34	48.26	**10.42**	**22.92**	43.75
	3-shot	**49.87**	**50.05**	**49.86**	**50.00**	6.25	20.83	**47.92**
Claude Fable 5.1	0-shot	39.35	39.53	39.19	39.24	0.00	8.33	41.67
	1-shot	40.55	40.30	40.26	40.28	6.25	2.08	37.50
	2-shot	41.34	41.58	41.27	41.32	6.25	12.50	37.50
	3-shot	41.75	41.76	41.64	41.67	2.08	8.33	39.58
Random and human references								
Uniform random	Expected	25.00	25.00	24.93	25.00	1.56	6.25	6.25
Human (reported)	Sampled	98.56	98.40	98.48	98.40	–	–	–

Model P/R/F1 are macro-averaged over A–D. GEM denotes FQ1-GEM. Bold marks model-column maxima. LoRA uses one checkpoint per family. Invalid answers count as incorrect. Random gives finite-test expectations for independent uniform A–D guesses. Human reports one participant's sampled blind evaluation, as summarized by the authors. Dashes denote unreported joint scores. Appendix A details the reference rows.

Table 4: All-images minus text-only in percentage points, with questions and gold labels fixed.

Model	ΔAcc.	ΔNTS	ΔDIS
Qwen2.5-VL-7B	−3.1	+2.1	+43.8
Qwen2.5-VL-7B LoRA	+3.1	+12.5	+12.5
InternVL3-8B	−1.4	+8.3	+12.5
InternVL3-8B LoRA	+2.4	+8.3	+8.3
Qwen3-VL-8B	+4.5	+14.6	+33.3
LLaVA-OneVision-7B	+0.0	+0.0	−6.2
MiniCPM-V 4.5	+3.8	+4.2	+14.6
Mistral Small 3.2 24B	−3.8	+2.1	+8.3
Qwen2.5-VL-32B	+2.4	+6.2	+35.4
InternVL3.5-14B	−2.1	+6.2	+10.4
Phi-4-Multimodal	−2.4	+0.0	+16.7
Kimi-VL-A3B	+4.2	+0.0	+4.2
GPT-5.6 / 0-shot	**+8.7**	+14.6	+35.4

Positive values favor all-images. Unmarked open-weight rows are Base. Each row compares the same 288 questions, with 48 groups per joint metric. LoRA rows use the primary adapters. Appendix D reports paired intervals for open-weight models.

below zero under exact-image, scene and union-component clustering. By contrast, NTS intervals include zero under the latter two schemes, as shown in Appendix D.6.

6.4 Option-Order Robustness

Following MirrorQA (Liu et al., 2026), we apply four cyclic option rotations while preserving images and questions.

Vanilla scores rotation zero. Circular requires four correct answers. Consistency requires four valid answers with identical decoded meanings, whether correct or incorrect.

Table 5b distinguishes stability from correctness. GPT leads Circular at **18.75%**, whereas Qwen LoRA is more consistent but reaches **16.20±0.53%** Circular accuracy. InternVL's higher Vanilla accuracy after adaptation accompanies lower Circular accuracy and consistency.

Responses also vary at fixed option order. A repeated GPT run reaches 45.14% accuracy versus 45.49% for rotation zero, but changes 46.53% of answers. Two-run joint correctness is 28.47% at fixed order and 28.12–31.25% across rotations in Table 6. Their similar rates show that rotation scores also reflect execution variability, as the paired analysis in Appendix A shows.

Table 5: Ordinary and Circular evaluation (%) on 288 questions. Open-weight controls are matched within panels. LoRA gives three-seed mean ± sample SD. GPT is zero-shot.

(a) Ordinary all-images evaluation

Setting	Acc.	GEM	NTS	DIS
InternVL3-8B				
Base	30.21	2.08	8.33	20.83
LoRA	32.41±1.31	0.69±1.20	9.72±1.20	12.50±2.08
Qwen2.5-VL-7B				
Base	29.17	0.00	2.08	45.83
LoRA	36.00±1.40	0.00±0.00	10.42±3.61	12.50±0.00
GPT-5.6				
0-shot	40.62	4.17	14.58	35.42

(b) Four option rotations

Setting	Vanilla	Circular	Consistency
InternVL3-8B			
Base	30.21	7.64	23.61
LoRA	32.75±1.31	6.02±1.12	17.59±1.57
Qwen2.5-VL-7B			
Base (S)	14.93	2.08	5.90
LoRA	35.19±1.12	**16.20±0.53**	32.64±1.25
GPT-5.6			
0-shot	45.49	**18.75**	27.08

GEM is FQ1-GEM. Joint scores use 48 groups per family. Vanilla uses within-run predictions. Appendix A gives the source of the summary marked S and all per-seed metrics.

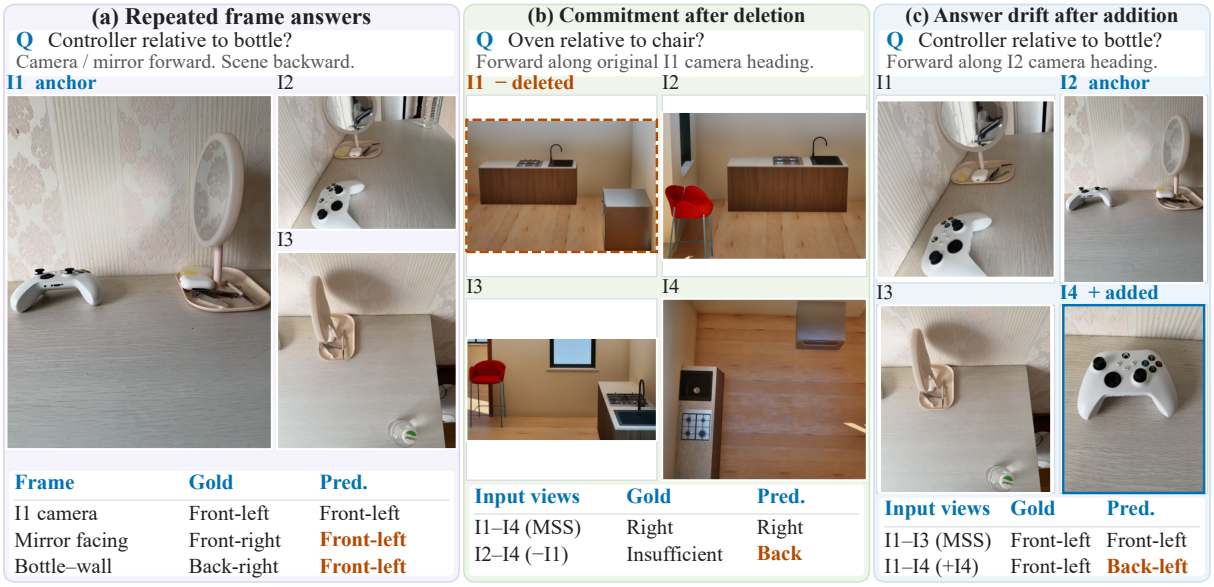


Figure 4: Selected GPT three-shot errors with released gold labels. I1–I4 mark original positions for display. Inputs use unlabeled images in their current order. Deletion in (b) removes the original camera anchor.

Table 6: GPT paired runs (%, $n = 288$). Nearly half of consistent pairs share a wrong answer.

Metric	Repeat	R_1	R_2	R_3
At least one correct	62.15	55.90	57.29	59.03
Both correct	28.47	31.25	30.56	28.12
Consistency	53.47	59.38	54.86	51.39
Consistently wrong	25.00	28.12	24.31	23.26

Columns pair R_0 with a same-order repeat or rotation. Consistency measures agreement in answer meaning.

7 Stratified Results and Failure Analysis

Answer changes need not be correct. Each FQ1 test triplet yields three frame pairs with different gold directions. Qwen3-VL-8B repeats a direction on 115/144 pairs, whereas GPT three-shot changes direction on 94/144 pairs but answers both correctly on only 27. Thus, models fail both to switch and to switch correctly. Figure 4a illustrates a repeated answer. Appendix F.2 compares all configurations.

From two to three shots, GPT's jointly correct pairs increase from 26 to 27 while fully correct triplets fall from five to three. Correct pairs are thus less concentrated in complete triplets, so pairwise gains need not improve group success.

Insufficiency responses extend beyond deletion. Both primary LoRA checkpoints increase insufficiency responses in all conditions in Table 7. Qwen switches 28 MSS and 22 addition answers from spatial directions to insufficiency, never the reverse. Deletion-label agreement rises while correctness on answerable inputs falls. Gateway models rarely choose insufficiency on MSS or addition inputs, whereas Mistral selects it more often on MSS than after deletion. These distributions distinguish selective abstention from general insufficiency bias.

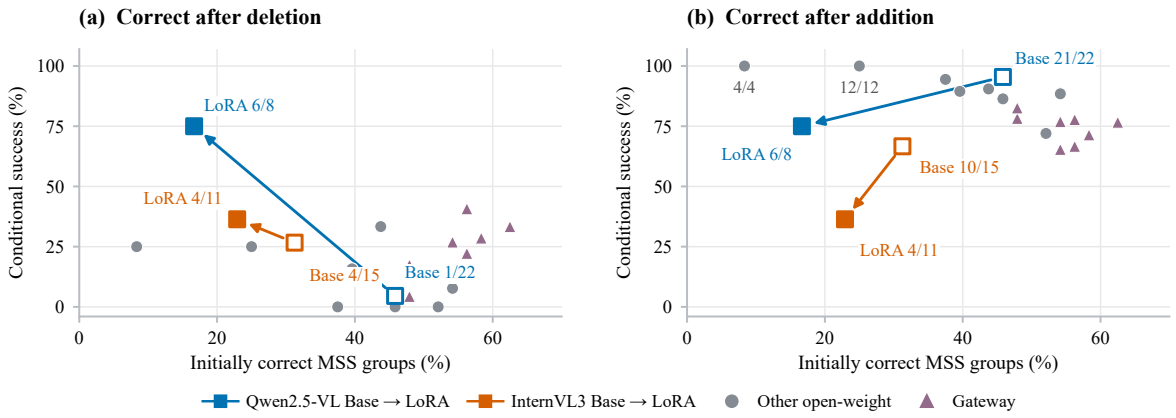


Figure 5: Initial and conditional success across 20 primary configurations. Horizontal rates use all 48 FQ2 groups. Vertical rates use each configuration's MSS-correct subset, with counts on highlighted points. Arrows link Base and LoRA, each with its own conditioning subset.

Table 7: Insufficiency counts out of 48.

Model / setting	MSS	Del.	Add.
Qwen2.5-VL-7B	2	3	2
InternVL3-8B	13	16	12
Qwen3-VL-8B	7	23	7
LLaVA-OneVision-7B	0	1	0
MiniCPM-V 4.5	0	6	0
Mistral Small 3.2 24B	29	24	21
Qwen2.5-VL-32B	9	18	8
InternVL3.5-14B	11	16	9
Phi-4-Multimodal	2	2	2
Kimi-VL-A3B	1	3	2
Qwen2.5-VL-7B LoRA	30	38	24
InternVL3-8B LoRA	22	30	21
GPT-5.6 / 0-shot	1	15	2
1-shot	1	14	2
2-shot	2	19	1
3-shot	2	18	1
Claude Fable 5.1 / 0-shot	1	10	0
1-shot	2	5	2
2-shot	3	15	3
3-shot	2	15	3

Unmarked open: Base. Del./Add.: deletion/addition. Counts /48 retain invalids: LLaVA addition 10; others 0.

MSS coverage separates intervention outcomes. On groups whose MSS both checkpoints solve, Qwen rises from 0/4 to 4/4 correct deletions, accounting for four of five net NTS gains in Figure 5. InternVL gains one shared-group deletion success, offset by changed MSS membership. Its joint deletion successes all involve removed original anchors, coupling evidence removal with reference changes.

The DIS declines have different sources. Qwen loses fourteen successes through changed MSS membership and one on shared groups. InternVL instead loses three through MSS membership changes and another three through addition errors within shared groups. Most of Qwen's decline thus originates before addition, whereas InternVL also loses robustness on groups both checkpoints solve at MSS. Appendix D.3 reports the decomposition.

Prompt effects vary across executions. Two randomized blocks on twelve FQ2 triplets excluding removed anchors yield 51.39% released-label agreement with original prompts and 52.78% with stable IDs. The near-equal item rates conceal lower joint agreement. Across 24 group–block observations, NTS falls from 8 to 4 and DIS from 9 to 8. Meanwhile, 14–15 of 36 answers change between blocks at fixed wording. Stable item rates therefore coexist with changes in which paired inputs receive correct labels. Appendix E describes the ID-based rewrite and its paired outcomes.

8 Conclusion

FrameEvidence evaluates whether spatial answers remain correct across matched changes in reference frames and visual evidence. Prompting and adaptation improve item accuracy without consistently improving joint correctness, while images most reliably benefit MSS–addition pairs. Comparing responses reveals why these outcomes diverge, from incorrect frame switches to changes in which MSS inputs a model solves. Repeated executions expose further instability beneath similar aggregate scores. Together, joint metrics and response comparisons distinguish individual gains from correctness across matched frame and evidence changes.

Limitations

FrameEvidence evaluates static, Chinese, four-option spatial reasoning on 96 held-out groups. Scenes can overlap across the image-separated splits, limiting conclusions about unseen-scene generalization. Metadata cues and image-ordinal changes can influence responses alongside visual evidence. Production review does not constitute independent response-level validation of labels or every-member MSS necessity. NTS therefore assesses agreement with the released deletion labels.

Model comparisons reflect the tested decoding settings, one demonstration pool and provider-reported gateway identities. The three-seed deviations describe training variation. Circular uses its own within-experiment controls. The human reference is one participant's sampled summary without individual answers or a documented sample size.

Ethical Considerations

The captured and rendered images were supplied by an external data provider with permission for public release. Language-model assistance supported editing, scripting and simulated review. The authors remain responsible for checking the evidence, citations and privacy of the released materials.

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A Implementation and Scoring Details

Evaluation inputs. Each public item contains the ordered image references and a Chinese prompt asking for exactly one of A, B, C or D. Evaluation identifiers link predictions to a private mapping with the source sample and group, input mode, gold letter and option meanings. Group identifiers and gold metadata are not shown to models. Metrics are computed from the frozen predictions and their matching input records. All model–mode comparisons use the full question and group counts.

Table 8: Model repositories and API aliases used in evaluation. Repositories identify model families. GPT and Claude names are provider-reported gateway aliases.

Displayed name	Configured repository or gateway identifier
Qwen2.5-VL-7B	`Qwen/Qwen2.5-VL-7B-Instruct`
InternVL3-8B	`OpenGVLab/InternVL3-8B-hf`
Qwen3-VL-8B	`Qwen/Qwen3-VL-8B-Instruct`
LLaVA-OneVision-7B	`llava-hf/llava-onevision-qwen2-7b-ov-hf`
MiniCPM-V 4.5	`openbmb/MiniCPM-V-4_5`
Mistral Small 3.2 24B	`mistralai/Mistral-Small-3.2-24B-Instruct-2506`
Qwen2.5-VL-32B	`Qwen/Qwen2.5-VL-32B-Instruct`
InternVL3.5-14B	`OpenGVLab/InternVL3_5-14B-HF`
Phi-4-Multimodal	`microsoft/Phi-4-multimodal-instruct`
Kimi-VL-A3B	`moonshotai/Kimi-VL-A3B-Instruct`
GPT-5.6	`gpt-5.6-sol`
Claude Fable 5.1	`claude-fable-5-1`

Offline scoring companion. A self-contained Python standard-library scorer reproduces the 20 primary configurations in Table 3, eight ordinary multi-seed configurations in Table 10, and seven complete Circular rows in Table 11. It uses sanitized per-item records without model calls or image access, preserving all item and group denominators. Primary, multi-seed and rotation runs remain separate. It also reproduces GPT's 1,152 Circular responses and 288 same-order repeat responses, including all 16 entries in Table 6. Running `python run_reproduction.py` from its `offline_reproduction` directory writes the recomputed tables and a verification report. The author-confirmed Qwen Base summary is reported separately from these recomputed results.

Production review and release correspondence. The production database contains 323 PASS and three REWORK actions. Selecting the latest PASS per group covers all 320 groups and 960 questions. Distinct annotation and review account assignments are recorded for every group, with semantic answers visible in the review interface. These are production approvals rather than independently collected answer sheets. Automatic comments and checklists do not verify reviewer independence or the answerability of every single-view deletion.

The task records exactly match the released annotations. Compared with the latest PASS snapshots, the release preserves all 960 semantic gold answers and option-meaning sets, and all 1,120 reviewed image files match. Raw fields nevertheless differ in 917 questions. Of 857 prompt changes, 856 replace a recurring Chinese-character typo and one revises a training item's anchor (`fce_test_00581`). Option order changes in 540 items and answer letters in 416, without changing semantic gold. Two anchor fields also change. Author adjudication confirms the scored inputs for the four flagged records.

Input and assembly consistency. A separate audit compares literal questions, ordered source-image bytes and options across all 960 items. It finds 959 distinct input signatures. The only repeated signature belongs to two training items with the same gold answer. No signature crosses splits or conflicts in its gold letter. All 288 test inputs are distinct. Ignoring option placement alone also yields no conflicting selected option strings.

All 320 triplets pass the encoded assembly rules. FQ1 preserves ordered images, options and entity fields across its three frame variants and has at least two distinct gold meanings. FQ2 preserves question and option strings, removes or adds exactly one image, and retains the order of the remaining images. No item repeats identical image content. Assets match production records. Geometric checks lack camera poses and object coordinates.

Few-shot construction. The fixed pool uses training groups `group_00152` for FQ1 and `group_00388` for FQ2. FQ1 demonstrations follow camera, object and scene order. FQ2 demonstrations follow MSS, deletion and addition order. One-, two- and three-shot conditions take the corresponding prefixes. The manifest verifies group and exact-image separation from test, although physical scenes may overlap. All shot counts use this fixed demonstration pool.

Decoding and invalid answers. The common output parser extracts an allowed answer letter and records the raw response, parsing status and API failure separately. An invalid response is never converted to a guessed answer. In class-wise recall and accuracy it contributes a false negative for its gold

class. Precision uses the predicted A–D counts, with zero for undefined per-class precision. Macro-F1 is the mean of the four classwise F1 scores.

A separate full-response parser recovers all 5,750 valid answers among 5,760 main-table records. The ten empty API failures remain incorrect. Reconstructed answers and point estimates match the frozen results.

Random and human reference rows. Uniform random selects one of A–D independently for each question. The reference row gives exact finite-test expectations. Accuracy and macro-recall are 25%, while macro-F1 averages the expected classwise ratios under the actual supports 75, 70, 69 and 74, yielding 24.93%. The three-answer and two-answer joint expectations are respectively $(1/4)^3$ and $(1/4)^2$. Separate binomial calculations and small-test enumeration verify the scorer.

The human reference is an author-reported P/R/F1/Acc. summary from one participant answering sampled groups. The retained record contains aggregate scores without the sample size or item-level responses. We report these sample-level values and leave joint scores unreported.

Held-out metadata controls. The post-hoc controls in Table 9 fix three categorical majority rules before computing their held-out scores. They fit only on the 288 FQ2 training questions in 96 groups. Neither the 48 development questions nor the 144 test questions select rules or features. Ties use alphabetic letter order, and unseen feature values fall back to the training-global majority. No held-out feature is unseen. The fixed features are image count alone and its pairing with the insufficiency option's letter, as in the original audit.

The count-plus-schema rule answers 54/144 test questions correctly but solves only 3/48 MSS/deletion pairs and 1/48 MSS/addition pairs. Conversely, always predicting the training-majority letter solves 11/48 addition pairs because the spatial answer letter is shared within each pair. Joint correctness therefore reveals differences between these rules beyond item accuracy, while the image-free control shows that shared answer preferences can also yield joint success. The reproduction materials include fitted rules and predictions.

Adaptation configuration. The LoRA checkpoints in Table 3 belong to the primary comparison. The ordinary multi-seed experiment in Table 10 evaluates two matched base controls and six independently trained adapters. Table 5 and Table 11 use those adapted checkpoints in separate rotation runs. All three experiments use the same 288 test questions but separate predictions. Vanilla uses Circular's own rotation-zero predictions.

Training uses LoRA rank 32, alpha 16 and zero dropout, with the vision tower and multimodal projector frozen. It specifies a 4,096-token cutoff, learning rate 2×10^{-5}, six epochs, cosine schedule, warmup ratio 0.05, and batch size one per device with eight accumulation steps on two devices. This gives effective batch size 16. The training export contains 576 train and 96 development items and excludes test responses. Seeds 20260722–20260724 are used in the three-seed experiment. All six runs complete 216 optimization steps over six epochs. Development loss selects step 180 for Qwen seeds 20260722 and 20260723, and step 216 for the remaining runs. Training and development exports contain no test sample identifiers. The companion reproduces scores from saved predictions and records the training configuration. Circular records contain four option rotations per source item. Their prompts and image order match the ordinary inputs at rotation zero. Separate executions nevertheless disagree on some items, so Vanilla uses the rotation run's own predictions.

Circular metrics. Let a_{ir} be the prediction for source item i at rotation r, mapped back to the original option identity, and let y_i be its gold identity. For each item, let $C_i = \prod_{r=0}^{3} \mathbb{1}[a_{ir} = y_i]$ indicate correctness across all four rotations. Let $S_i = \mathbb{1}[a_{i0} = a_{i1} = a_{i2} = a_{i3}]$ indicate consistency. With $N = 288$, we compute

$$\text{Circular} = \frac{\sum_i C_i}{N}, \quad \text{Cons.} = \frac{\sum_i S_i}{N}. \qquad (2)$$

All four predictions must be valid for consistency to receive credit. The reference implementation additionally reports each rotation and the mean over rotations. Table 11 retains the per-seed scores. Family means and sample deviations are calculated from unrounded item counts. All six adapted checkpoints and the InternVL Base control have complete retained predictions. Qwen Base is reported from the final run summary rather than recomputed from the retained prediction copy. The GPT supplement adds 1,152 valid zero-shot responses over the same 288 source questions. Its four rotations yield 131, 120, 122 and 120 correct answers. There are 54 fully correct sets and 78 semantically consistent

Table 9: Train-fitted FQ2 metadata controls (%). Fixed categorical majority rules use 288 training questions. Dev and test contain 48 and 144 questions, respectively. Joint test metrics use all 48 FQ2 groups. No pixels, question text or condition labels are used.

Input features	Train Acc.	Dev Acc.	Test Acc.	Test NTS	Test DIS
Majority letter	26.39	20.83	23.61	0.00	22.92
Image count	29.51	12.50	21.53	0.00	4.17
Count + insufficiency letter	47.92	43.75	37.50	6.25	2.08

Table 10: Ordinary all-images multi-seed results (%). GEM denotes FQ1-GEM. Classification uses 288 questions and joint criteria use 48 groups per family. Means and sample deviations use three LoRA seeds. These ordinary runs are separate from Circular rotation zero.

Family	Run	P	R	F1	Acc.	GEM	NTS	DIS
InternVL3-8B	Base	30.34	30.54	29.36	30.21	2.08	8.33	20.83
	20260722	34.32	33.20	31.50	32.99	0.00	10.42	14.58
	20260723	32.09	31.19	29.11	30.90	0.00	8.33	10.42
	20260724	33.85	33.64	31.85	33.33	2.08	10.42	12.50
	Mean	33.42±1.17	32.68±1.31	30.82±1.49	32.41±1.31	0.69±1.20	9.72±1.20	12.50±2.08
Qwen2.5-VL-7B	Base	30.87	29.77	26.35	29.17	0.00	2.08	45.83
	20260722	35.74	35.88	35.00	35.76	0.00	8.33	12.50
	20260723	35.10	34.85	33.99	34.72	0.00	8.33	12.50
	20260724	37.53	37.57	37.10	37.50	0.00	14.58	12.50
	Mean	36.12±1.26	36.10±1.37	35.36±1.59	36.00±1.40	0.00±0.00	10.42±3.61	12.50±0.00

sets. Its Vanilla count is therefore 131/288, versus 117/288 in the primary all-images run.

Qwen Base's rounded summary also matches scores from the incomplete copy when its 446 failed calls count as incorrect. That copy cannot independently confirm completion. We retain the author-confirmed summary as a reference in Tables 5 and 11. It is separate from the independently recomputed paired estimates.

GPT cross-run stability. Pairing ordinary all-images and Circular rotation zero on all 288 source items gives 79 correct in both executions, 38 correct only in ordinary, 52 correct only in rotation zero, and 119 incorrect in both. Their answer letters differ on 136 items (47.22%). Prepared prompts, option order, ordered image references and retained decoding configurations match. Archived wire requests and backend revisions are unavailable, so this is a descriptive cross-run comparison, not an isolated decoding experiment. We resample whole clusters with replacement 20,000 times, using seed 20261010 and item-weighted ratios. Across the 13 union components defined in Appendix D.6, the 95% percentile interval for disagreement is [41.33, 56.61]%. The accuracy difference is 4.86 points with interval [−3.67, 16.67]. These intervals quantify sensitivity to item composition under this clustering, not repeated API execution.

Same-order control. We repeat the same 288 zero-shot all-images queries with identical GPT request settings. All responses are valid. The repeat yields 130 correct answers, with P/R/F1/Acc of 45.66/45.31/45.06/45.14%. Its FQ1-GEM, NTS and DIS are 4.17%, 20.83% and 31.25%. Paired with the primary run, 77 answers are correct in both, 40 only in the primary, 53 only in the repeat, and 118 in neither. Predictions change on 128/288 items. Pairing with rotation zero gives counts of 82, 49, 48 and 109, with 134 changed answers.

Table 6 compares two-answer sets sharing the same rotation-zero output. The same-order pair is jointly correct on 82 items and semantically consistent on 154. Pairing rotation zero with rotations one, two and three gives joint counts of 90, 88 and 81 and consistency counts of 171, 158 and 148. Holding set size fixed removes the two-versus-four conjunction difference when assessing stability.

We resample paired observations in 20,000 whole-cluster bootstrap draws with seed 20261010, using both 96 triplets and 13 image/scene union components. Table 12 reports estimates with both 95% intervals. Every joint-correctness contrast with a rotation pair includes zero under both schemes. The consistency contrast with rotation one is below zero under union clustering but includes zero under triplet clustering. These intervals describe question-composition sensitivity. The repeat and rotation runs were executed at different

Table 11: Per-seed open-weight and zero-shot GPT Circular scores (%) for 288 items and four cyclic rotations. Drop uses percentage points. Qwen Base is summary-reported. Other rows use 1,152 retained valid responses each.

Family	Seed / control	Vanilla	Mean rot.	Circular	Consistency	Drop
InternVL3-8B	Base	30.21	28.04	7.64	23.61	22.57
	20260722	33.33	31.42	7.29	19.10	26.04
	20260723	33.68	31.08	5.56	17.71	28.12
	20260724	31.25	30.30	5.21	15.97	26.04
Qwen2.5-VL-7B	Base	14.93	18.75	2.08	5.90	12.85
	20260722	34.38	35.76	16.67	33.68	17.71
	20260723	34.72	35.42	15.62	32.99	19.10
	20260724	36.46	36.81	16.32	31.25	20.14
GPT-5.6	0-shot	45.49	42.80	18.75	27.08	26.74

Table 12: Paired GPT same-order control with 95% bootstrap intervals. Rates are percentages and differences are percentage points. Rotation contrasts subtract the corresponding R_0–R_j pair from the R_0–repeat pair.

Quantity	Estimate	96 triplets	13 union components
Repeat accuracy	45.14	[39.58, 50.69]	[39.09, 50.56]
Primary–repeat disagreement	44.44	[38.54, 50.69]	[37.92, 54.49]
R_0–repeat disagreement	46.53	[40.28, 52.78]	[43.18, 51.96]
Repeat minus primary accuracy	4.51	[−1.74, 10.76]	[−1.99, 14.29]
Repeat minus R_0 accuracy	−0.35	[−6.94, 6.25]	[−12.28, 6.98]
Same-order pair both correct	28.47	[23.26, 33.68]	[23.50, 33.90]
Same-order pair consistency	53.47	[47.22, 59.72]	[48.04, 56.82]
Both correct, difference from R_0–R_1	−2.78	[−7.29, 1.39]	[−10.12, 1.55]
Both correct, difference from R_0–R_2	−2.08	[−6.60, 2.43]	[−8.84, 3.03]
Both correct, difference from R_0–R_3	0.35	[−4.17, 4.86]	[−7.21, 5.51]
Consistency, difference from R_0–R_1	−5.90	[−13.19, 1.04]	[−15.33, −0.65]
Consistency, difference from R_0–R_2	−1.39	[−8.33, 5.56]	[−9.60, 7.34]
Consistency, difference from R_0–R_3	2.08	[−4.51, 8.33]	[−7.04, 7.68]

times, so temporal service changes and decoding variation remain combined.

B Release Composition

Figure 6 shows condition balance and source object-label frequencies, counted once per group. The Other category aggregates labels outside the ten most frequent types.

C Complete Input Controls

Table 13 includes twelve open-weight configurations and GPT zero-shot. Original gold labels remain fixed even when an ablation removes information needed to infer them. GPT retains 288 valid responses per mode and uses the primary table's all-images predictions. Its companion provides sanitized per-item inputs, predictions and exact counts.

D Stratified Results and Paired Contrasts

The following tables report all-images results across size and scene strata. Each stratum contains different instances, making these comparisons descriptive. The FQ2 frame strata retain annotation tags rather than independently validated frame categories. A lexical audit flags 30 of 48 MSS prompts that explicitly name an image's camera heading despite a non-camera tag. These comprise 14 of 16 object-tagged groups and all 16 world-tagged groups. The partitions use annotation tags and flag explicit camera wording separately.

D.1 Geometry-associated differences

Table 14 shows that axial relations favor DIS across most configurations, but NTS has no median gap. Granularity is associated more consistently with addition than deletion-label success. Shared scene-dependent groups preclude treating configurations as independent replications.

D.2 Unconditional response distributions

Table 7 counts insufficient-information responses in every condition, with all 48 groups retained. Option meanings, not letters, define this category. All 576 FQ2 predictions from the four primary Base/LoRA checkpoints are valid. For Qwen, the Base-to-LoRA spatial-to-insufficiency switches number 28, 35 and 22 for MSS, deletion and ad-

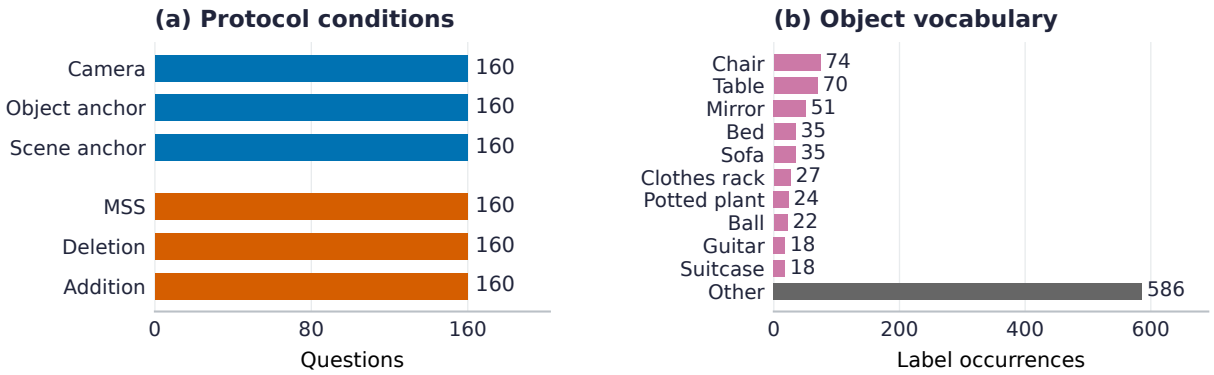


Figure 6: Release composition. Each of six conditions contains 160 questions. Object-label occurrences are counted once per matched group across 126 distinct labels. Other pools labels outside the top ten.

Table 13: Visual-input ablations on 288 test questions (%). Questions, options and gold answers are held fixed across all-images, first-image and text-only modes.

Model	All images				First image				Text only			
	Acc.	GEM	NTS	DIS	Acc.	GEM	NTS	DIS	Acc.	GEM	NTS	DIS
Qwen2.5-VL-7B Base	28.82	0.00	2.08	43.75	29.17	0.00	2.08	37.50	31.94	0.00	0.00	0.00
Qwen2.5-VL-7B LoRA	35.42	0.00	12.50	12.50	31.25	0.00	0.00	0.00	32.29	2.08	0.00	0.00
InternVL3-8B Base	29.51	2.08	8.33	20.83	27.78	0.00	6.25	29.17	30.90	2.08	0.00	8.33
InternVL3-8B LoRA	32.64	0.00	8.33	8.33	27.78	0.00	4.17	10.42	30.21	0.00	0.00	0.00
Qwen3-VL-8B Base	34.03	0.00	14.58	39.58	28.82	0.00	8.33	10.42	29.51	0.00	0.00	6.25
LLaVA-OneVision-7B Base	29.17	0.00	0.00	37.50	31.60	0.00	0.00	50.00	29.17	0.00	0.00	43.75
MiniCPM-V 4.5 Base	32.99	0.00	4.17	47.92	31.94	0.00	8.33	43.75	29.17	0.00	0.00	33.33
Mistral Small 3.2 24B Base	25.69	0.00	2.08	8.33	29.51	0.00	6.25	31.25	29.51	0.00	0.00	0.00
Qwen2.5-VL-32B Base	33.33	0.00	6.25	35.42	26.74	0.00	8.33	16.67	30.90	0.00	0.00	0.00
InternVL3.5-14B Base	28.12	0.00	6.25	25.00	24.65	0.00	8.33	8.33	30.21	0.00	0.00	14.58
Phi-4-Multimodal Base	26.39	0.00	0.00	35.42	28.12	0.00	4.17	37.50	28.82	2.08	0.00	18.75
Kimi-VL-A3B Base	29.17	0.00	0.00	39.58	30.21	2.08	6.25	31.25	25.00	0.00	0.00	35.42
GPT-5.6 / 0-shot	40.62	4.17	14.58	35.42	29.51	0.00	12.50	8.33	31.94	0.00	0.00	0.00

Table 14: Axial–quadrant gaps over 20 configurations (24 groups per stratum). Signs count positive, tied and negative gaps. Median Δ is in percentage points.

Metric	$>$	$=$	$<$	Median Δ
FQ1-GEM	2	13	5	0.0
NTS	5	8	7	0.0
DIS	17	3	0	+20.8

dition, with no reverse switches. InternVL has 10, 14 and nine such switches, with one reverse switch in MSS and none elsewhere. The paired answers reveal a broader shift toward insufficiency across all three conditions.

Conditional failures. Conditional deletion and addition outcomes are computed only where the MSS answer was initially correct. Their denominators therefore differ by model. Unconditional NTS and DIS instead always use all 48 FQ2 groups. Keeping both views avoids attributing low joint success entirely to the intervention when the base question was already answered incorrectly.

D.3 Common initially correct groups

To separate response changes from changes in conditional-set membership, we partition the 48 FQ2 groups by the primary Base and LoRA checkpoints' MSS correctness. The both-correct, Base-only, LoRA-only and neither-correct counts are 4/18/4/22 for Qwen and 9/6/2/31 for InternVL. Table 20 compares intervention answers on the common groups and reconstructs each full joint difference. On those groups, deletion has four gains and no losses for Qwen, versus one gain and no losses for InternVL. Addition loses one Qwen and three InternVL successes, with no gains.

The remaining contribution comes from intervention successes in groups whose MSS only one checkpoint answers correctly. Groups that neither answers correctly contribute zero to both joint scores, whatever their intervention answers. This exactly decomposes the original 48-group numerators. The common sets are small and selected by observed outcomes, so the paired changes describe

Table 15: Evidence-set size strata (%). Each column gives its group denominator n.

Model	FQ1-GEM			NTS			DIS		
	2 $n=15$	3 $n=16$	4 $n=17$	2 $n=17$	3 $n=16$	4 $n=15$	2 $n=17$	3 $n=16$	4 $n=15$
Claude Fable 5.1 (0-shot)	0.0	0.0	0.0	11.8	12.5	0.0	23.5	56.2	46.7
Claude Fable 5.1 (1-shot)	6.7	12.5	0.0	0.0	6.2	0.0	29.4	37.5	46.7
Claude Fable 5.1 (2-shot)	0.0	12.5	5.9	17.6	18.8	0.0	35.3	37.5	40.0
Claude Fable 5.1 (3-shot)	0.0	6.2	0.0	11.8	12.5	0.0	29.4	56.2	33.3
GPT-5.6 (0-shot)	6.7	0.0	5.9	17.6	25.0	0.0	35.3	25.0	46.7
GPT-5.6 (1-shot)	0.0	0.0	17.6	11.8	31.2	6.7	41.2	43.8	40.0
GPT-5.6 (2-shot)	0.0	12.5	17.6	23.5	43.8	0.0	41.2	56.2	33.3
GPT-5.6 (3-shot)	6.7	12.5	0.0	29.4	25.0	6.7	41.2	50.0	53.3
InternVL3-8B Base	6.7	0.0	0.0	5.9	6.2	13.3	17.6	25.0	20.0
InternVL3-8B LoRA	0.0	0.0	0.0	11.8	6.2	6.7	11.8	12.5	0.0
InternVL3.5-14B	0.0	0.0	0.0	5.9	6.2	6.7	17.6	18.8	40.0
Kimi-VL-A3B	0.0	0.0	0.0	0.0	0.0	0.0	41.2	43.8	33.3
LLaVA-OneVision-7B	0.0	0.0	0.0	0.0	0.0	0.0	41.2	43.8	26.7
MiniCPM-V 4.5	0.0	0.0	0.0	5.9	6.2	0.0	41.2	37.5	66.7
Mistral Small 3.2 24B	0.0	0.0	0.0	5.9	0.0	0.0	11.8	0.0	13.3
Phi-4-Multimodal	0.0	0.0	0.0	0.0	0.0	0.0	29.4	31.2	46.7
Qwen2.5-VL-32B	0.0	0.0	0.0	5.9	12.5	0.0	35.3	31.2	40.0
Qwen2.5-VL-7B Base	0.0	0.0	0.0	5.9	0.0	0.0	47.1	56.2	26.7
Qwen2.5-VL-7B LoRA	0.0	0.0	0.0	0.0	25.0	13.3	0.0	25.0	13.3
Qwen3-VL-8B	0.0	0.0	0.0	23.5	12.5	6.7	29.4	37.5	53.3

All configurations use the all-images test, with strata comparing different held-out groups.

Table 16: Axial and quadrant relation strata (%). Each column gives its group denominator n.

Model	FQ1-GEM		NTS		DIS	
	Axial $n=24$	Quadrant $n=24$	Axial $n=24$	Quadrant $n=24$	Axial $n=24$	Quadrant $n=24$
Claude Fable 5.1 (0-shot)	0.0	0.0	16.7	0.0	54.2	29.2
Claude Fable 5.1 (1-shot)	4.2	8.3	0.0	4.2	45.8	29.2
Claude Fable 5.1 (2-shot)	4.2	8.3	12.5	12.5	54.2	20.8
Claude Fable 5.1 (3-shot)	0.0	4.2	8.3	8.3	50.0	29.2
GPT-5.6 (0-shot)	4.2	4.2	16.7	12.5	45.8	25.0
GPT-5.6 (1-shot)	8.3	4.2	8.3	25.0	41.7	41.7
GPT-5.6 (2-shot)	8.3	12.5	33.3	12.5	50.0	37.5
GPT-5.6 (3-shot)	4.2	8.3	25.0	16.7	58.3	37.5
InternVL3-8B Base	4.2	0.0	8.3	8.3	29.2	12.5
InternVL3-8B LoRA	0.0	0.0	8.3	8.3	12.5	4.2
InternVL3.5-14B	0.0	0.0	8.3	4.2	37.5	12.5
Kimi-VL-A3B	0.0	0.0	0.0	0.0	66.7	12.5
LLaVA-OneVision-7B	0.0	0.0	0.0	0.0	50.0	25.0
MiniCPM-V 4.5	0.0	0.0	0.0	8.3	58.3	37.5
Mistral Small 3.2 24B	0.0	0.0	0.0	4.2	8.3	8.3
Phi-4-Multimodal	0.0	0.0	0.0	0.0	54.2	16.7
Qwen2.5-VL-32B	0.0	0.0	4.2	8.3	50.0	20.8
Qwen2.5-VL-7B Base	0.0	0.0	0.0	4.2	50.0	37.5
Qwen2.5-VL-7B LoRA	0.0	0.0	12.5	12.5	12.5	12.5
Qwen3-VL-8B	0.0	0.0	12.5	16.7	50.0	29.2

All configurations use the all-images test, with strata comparing different held-out groups.

these subsets rather than causal adaptation effects. Reproduction materials retain group assignments and paired outcomes.

D.4 Fixed-marginal joint-score diagnostic

Low conjunction rates can follow from low component accuracy. Independent uniform guesses give expected rates of $1/64$ for FQ1-GEM and $1/16$ for NTS and DIS, but ignore the observed variant accuracies. We therefore add a descriptive reference that keeps each variant's correct count fixed. Let C_{gj} indicate correctness for variant j in group g, with $c_j = \sum_g C_{gj}$ and $n = 48$. The observed joint count is $J = \sum_g \prod_{j=1}^{k} C_{gj}$. Fixing one correctness column and uniformly, independently rearranging the others across groups gives the exact reference rate $R = \prod_{j=1}^{k}(c_j/n)$. No sampled per-

Table 17: Scene-category strata (%). Each column gives its group denominator n.

Model	FQ1-GEM			NTS			DIS		
	Large $n = 15$	Tabletop $n = 17$	Outdoor $n = 16$	Large $n = 16$	Tabletop $n = 17$	Outdoor $n = 15$	Large $n = 16$	Tabletop $n = 17$	Outdoor $n = 15$
Claude Fable 5.1 (0-shot)	0.0	0.0	0.0	12.5	5.9	6.7	50.0	23.5	53.3
Claude Fable 5.1 (1-shot)	13.3	5.9	0.0	6.2	0.0	0.0	31.2	41.2	40.0
Claude Fable 5.1 (2-shot)	6.7	5.9	6.2	12.5	11.8	13.3	25.0	35.3	53.3
Claude Fable 5.1 (3-shot)	0.0	5.9	0.0	18.8	5.9	0.0	37.5	35.3	46.7
GPT-5.6 (0-shot)	0.0	11.8	0.0	18.8	17.6	6.7	37.5	41.2	26.7
GPT-5.6 (1-shot)	6.7	5.9	6.2	31.2	0.0	20.0	56.2	29.4	40.0
GPT-5.6 (2-shot)	13.3	17.6	0.0	50.0	5.9	13.3	75.0	17.6	40.0
GPT-5.6 (3-shot)	0.0	5.9	12.5	31.2	17.6	13.3	56.2	47.1	40.0
InternVL3-8B Base	0.0	5.9	0.0	6.2	5.9	13.3	37.5	11.8	13.3
InternVL3-8B LoRA	0.0	0.0	0.0	12.5	11.8	0.0	6.2	11.8	6.7
InternVL3.5-14B	0.0	0.0	0.0	6.2	5.9	6.7	25.0	23.5	26.7
Kimi-VL-A3B	0.0	0.0	0.0	0.0	0.0	0.0	31.2	29.4	60.0
LLaVA-OneVision-7B	0.0	0.0	0.0	0.0	0.0	0.0	25.0	29.4	60.0
MiniCPM-V 4.5	0.0	0.0	0.0	6.2	5.9	0.0	43.8	41.2	60.0
Mistral Small 3.2 24B	0.0	0.0	0.0	0.0	0.0	6.7	0.0	17.6	6.7
Phi-4-Multimodal	0.0	0.0	0.0	0.0	0.0	0.0	31.2	29.4	46.7
Qwen2.5-VL-32B	0.0	0.0	0.0	6.2	11.8	0.0	25.0	47.1	33.3
Qwen2.5-VL-7B Base	0.0	0.0	0.0	0.0	0.0	6.7	31.2	41.2	60.0
Qwen2.5-VL-7B LoRA	0.0	0.0	0.0	12.5	11.8	13.3	25.0	5.9	6.7
Qwen3-VL-8B	0.0	0.0	0.0	18.8	11.8	13.3	37.5	41.2	40.0

All configurations use the all-images test, with strata comparing different held-out groups.

Table 18: FQ2 annotation-tag strata (%). Each column gives its group denominator n.

Model	NTS			DIS		
	Camera tag $n = 16$	Object tag $n = 16$	World tag $n = 16$	Camera tag $n = 16$	Object tag $n = 16$	World tag $n = 16$
Claude Fable 5.1 (0-shot)	12.5	6.2	6.2	43.8	37.5	43.8
Claude Fable 5.1 (1-shot)	6.2	0.0	0.0	43.8	31.2	37.5
Claude Fable 5.1 (2-shot)	18.8	6.2	12.5	50.0	31.2	31.2
Claude Fable 5.1 (3-shot)	6.2	12.5	6.2	43.8	43.8	31.2
GPT-5.6 (0-shot)	12.5	12.5	18.8	37.5	31.2	37.5
GPT-5.6 (1-shot)	12.5	18.8	18.8	50.0	37.5	37.5
GPT-5.6 (2-shot)	25.0	12.5	31.2	50.0	37.5	43.8
GPT-5.6 (3-shot)	37.5	12.5	12.5	50.0	37.5	56.2
InternVL3-8B Base	6.2	12.5	6.2	12.5	31.2	18.8
InternVL3-8B LoRA	6.2	12.5	6.2	6.2	12.5	6.2
InternVL3.5-14B	12.5	6.2	0.0	31.2	37.5	6.2
Kimi-VL-A3B	0.0	0.0	0.0	31.2	43.8	43.8
LLaVA-OneVision-7B	0.0	0.0	0.0	50.0	25.0	37.5
MiniCPM-V 4.5	0.0	12.5	0.0	43.8	50.0	50.0
Mistral Small 3.2 24B	6.2	0.0	0.0	12.5	12.5	0.0
Phi-4-Multimodal	0.0	0.0	0.0	37.5	31.2	37.5
Qwen2.5-VL-32B	0.0	12.5	6.2	37.5	43.8	25.0
Qwen2.5-VL-7B Base	0.0	0.0	6.2	31.2	43.8	56.2
Qwen2.5-VL-7B LoRA	0.0	18.8	18.8	0.0	18.8	18.8
Qwen3-VL-8B	6.2	31.2	6.2	25.0	68.8	25.0

All-images test grouped by annotation tags. Thirty of 48 MSS prompts name a camera heading despite a non-camera tag. These tags have not been independently validated as frame categories.

mutations are needed. The possible joint count lies between $\max(0, \sum_j c_j - (k-1)n)$ and $\min_j c_j$.

Tables 21 and 22 retain all 20 primary and eight ordinary configurations separately. Primary NTS is below the reference in 19 rows, with gaps of -8.59 to $+0.65$ percentage points. DIS exceeds it in every row, with gaps of $+4.51$ to $+23.70$ points. FQ1-GEM has five positive and 15 negative gaps, so it does not support a uniform direction. All eight ordinary configurations have negative FQ1-GEM and NTS gaps and positive DIS gaps. These configurations share test groups and are not independent replications of a population-level effect.

The train-fitted controls in Table 23 make this limitation concrete. Always predicting the training majority letter B gives NTS of 0.00% against

Table 19: Failure profiles on the all-images test (%). Frame invariance is unchanged semantic output despite different gold directions.

Model	Frame invariant	m	Deletion given MSS correct			Addition		Invalid
			To $\perp$	Original	Other	Robust	Drift	
Claude Fable 5.1 (0-shot)	47.9	26	15.4	57.7	26.9	41.7	12.5	0.0
Claude Fable 5.1 (1-shot)	27.1	23	4.3	60.9	34.8	37.5	10.4	0.0
Claude Fable 5.1 (2-shot)	22.9	27	22.2	48.1	29.6	37.5	18.8	0.0
Claude Fable 5.1 (3-shot)	25.0	23	17.4	56.5	26.1	39.6	8.3	0.0
GPT-5.6 (0-shot)	20.8	26	26.9	42.3	30.8	35.4	18.8	0.0
GPT-5.6 (1-shot)	18.8	28	28.6	35.7	35.7	41.7	16.7	0.0
GPT-5.6 (2-shot)	18.8	27	40.7	29.6	29.6	43.8	12.5	0.0
GPT-5.6 (3-shot)	12.5	30	33.3	23.3	43.3	47.9	14.6	0.0
InternVL3-8B Base	50.0	15	26.7	60.0	13.3	20.8	10.4	0.0
InternVL3-8B LoRA	68.8	11	36.4	36.4	27.3	8.3	14.6	0.0
InternVL3.5-14B	66.7	12	25.0	66.7	8.3	25.0	0.0	0.0
Kimi-VL-A3B	62.5	22	0.0	86.4	13.6	39.6	6.2	0.0
LLaVA-OneVision-7B	70.8	25	0.0	80.0	20.0	37.5	2.1	20.8
MiniCPM-V 4.5	56.2	26	7.7	80.8	11.5	47.9	6.2	0.0
Mistral Small 3.2 24B	66.7	4	25.0	50.0	25.0	8.3	0.0	0.0
Phi-4-Multimodal	75.0	18	0.0	72.2	27.8	35.4	2.1	0.0
Qwen2.5-VL-32B	58.3	19	15.8	78.9	5.3	35.4	4.2	0.0
Qwen2.5-VL-7B Base	70.8	22	4.5	81.8	13.6	43.8	2.1	0.0
Qwen2.5-VL-7B LoRA	60.4	8	75.0	25.0	0.0	12.5	4.2	0.0
Qwen3-VL-8B	72.9	21	33.3	52.4	14.3	39.6	4.2	0.0

m counts correct MSS answers out of 48 and is the deletion denominator. Other percentages use 48 groups. Original/Other denote spatial answers after deletion.

Table 20: Initial Base–LoRA comparisons on common MSS-correct groups. The full joint-success change equals the common-group change plus LoRA-only successes minus Base-only successes. All entries are counts.

Family	Metric	Common groups Base $\rightarrow$ LoRA	Common change	LoRA-only successes	Base-only successes	Full change out of 48
Qwen2.5-VL-7B	NTS	$0/4 \rightarrow 4/4$	$+4$	2	1	$+5$
	DIS	$4/4 \rightarrow 3/4$	-1	3	17	-15
InternVL3-8B	NTS	$2/9 \rightarrow 3/9$	$+1$	1	2	0
	DIS	$7/9 \rightarrow 4/9$	-3	0	3	-6

a 5.73% reference and DIS of 22.92% against a 5.25% reference. Thus the same negative-NTS/positive-DIS pattern can arise without images. This follows from the shared spatial answer in the addition pair and its change to insufficiency after deletion. Each control retains its train-fitted rule.

The reference breaks the observed pairing while preserving variant accuracy. It therefore distinguishes how successes align from how often they occur. However, heterogeneous groups are not exchangeable by design. Shared difficulty and repeated answers can both affect the observed alignment. We use the gaps and feasible bounds as descriptive measures of this alignment. All 84 original joint counts are reproduced with all groups and invalid outputs retained. The companion supplies component counts and feasible bounds.

D.5 Uncertainty

The paired analysis resamples exact-image connected components as blocks while retaining paired outcomes within each contrast. It uses 10,000 bootstrap draws. These intervals condition on the held-out test and exact-image grouping. They do not include provider nondeterminism, demonstration-selection variance or all scene reuse. A zero-width interval at zero indicates no variation among observed groups, not zero population performance.

D.6 Clustering sensitivity

This post-hoc analysis compares four fixed LoRA–Base contrasts on 48 FQ2 groups. Table 25 reports partitions into 29 exact-image components and 14 inferred core-MSS scene/template clusters.

For each partition, 20,000 bootstrap draws (seed 20261009) sample its original number of clusters with replacement. Each sampled block retains all its paired groups. A replicate divides its signed

Table 21: Observed (O) and fixed-marginal rearrangement-reference (R) joint-success percentages for all primary configurations. Each criterion uses 48 groups. R preserves variant accuracy while independently rearranging answers across groups.

Configuration	FQ1-GEM		NTS		DIS	
	O	R	O	R	O	R
Qwen2.5-VL-7B (Base)	0.00	1.32	2.08	2.86	43.75	20.05
InternVL3-8B (Base)	2.08	1.88	8.33	10.42	20.83	8.46
Qwen3-VL-8B (Base)	0.00	0.80	14.58	20.96	39.58	17.32
LLaVA-OneVision-7B (Base)	0.00	1.11	0.00	1.09	37.50	19.53
MiniCPM-V 4.5 (Base)	0.00	0.98	4.17	6.77	47.92	28.21
Mistral Small 3.2 24B (Base)	0.00	1.56	2.08	4.17	8.33	1.74
Qwen2.5-VL-32B (Base)	0.00	1.14	6.25	14.84	35.42	17.32
InternVL3.5-14B (Base)	0.00	1.51	6.25	8.33	25.00	7.81
Phi-4-Multimodal (Base)	0.00	1.37	0.00	1.56	35.42	14.84
Kimi-VL-A3B (Base)	0.00	1.91	0.00	2.86	39.58	19.10
Qwen2.5-VL-7B (LoRA)	0.00	2.43	12.50	13.19	12.50	3.47
InternVL3-8B (LoRA)	0.00	2.79	8.33	14.32	8.33	3.82
GPT-5.6 (0-shot)	4.17	4.12	14.58	16.93	35.42	27.08
GPT-5.6 (1-shot)	6.25	5.53	16.67	17.01	41.67	35.24
GPT-5.6 (2-shot)	10.42	6.68	22.92	22.27	43.75	38.67
GPT-5.6 (3-shot)	6.25	8.23	20.83	23.44	47.92	40.36
Claude Fable 5.1 (0-shot)	0.00	2.40	8.33	11.28	41.67	31.60
Claude Fable 5.1 (1-shot)	6.25	6.79	2.08	4.99	37.50	23.96
Claude Fable 5.1 (2-shot)	6.25	3.26	12.50	17.58	37.50	28.12
Claude Fable 5.1 (3-shot)	2.08	4.48	8.33	14.97	39.58	24.96

Table 22: The same joint-score diagnostic for the separate ordinary multi-seed cohort. O denotes observed rates and R the fixed-marginal reference. Each row retains all 48 groups per criterion.

Configuration	FQ1-GEM		NTS		DIS	
	O	R	O	R	O	R
InternVL3-8B (matched control)	2.08	2.28	8.33	11.11	20.83	8.33
InternVL3-8B (LoRA 20260722)	0.00	2.63	10.42	15.10	14.58	5.73
InternVL3-8B (LoRA 20260723)	0.00	2.45	8.33	13.37	10.42	3.82
InternVL3-8B (LoRA 20260724)	2.08	3.01	10.42	13.89	12.50	3.91
Qwen2.5-VL-7B (matched control)	0.00	1.19	2.08	2.86	45.83	21.01
Qwen2.5-VL-7B (LoRA 20260722)	0.00	2.63	8.33	11.55	12.50	3.04
Qwen2.5-VL-7B (LoRA 20260723)	0.00	2.43	8.33	11.55	12.50	3.04
Qwen2.5-VL-7B (LoRA 20260724)	0.00	3.42	14.58	15.62	12.50	4.77

Table 23: The fixed-marginal diagnostic applied to all three retained metadata controls. O is observed joint success and R the rearrangement reference. Rules are fitted on training groups. The constant-letter control produces opposite NTS and DIS gaps using answer preferences alone.

Input features	Correct counts / 48			NTS (%)		DIS (%)	
	MSS	Del.	Add.	O	R	O	R
Majority letter	11	12	11	0.00	5.73	22.92	5.25
Image count	9	9	13	0.00	3.52	4.17	5.08
Count + insufficiency letter	5	42	7	6.25	9.11	2.08	1.52

success difference by its sampled group count, accounting for unequal block sizes. The original point estimates always divide by 48.

These partitions are not nested. One exact-image component spanning groups 00360, 00361 and 00364 crosses two inferred scene clusters. We therefore also join groups connected by either relation, including indirect paths. This union yields 13 components, with at most 15 groups each. We apply the same bootstrap scheme and independently verify all 20,000 draws. Neither scene labels nor union components certify independent physical scenes. Both NTS intervals include zero, while both DIS intervals remain negative under all three partitions. InternVL's unchanged NTS also masks two newly successful and two newly failed groups.

For a cluster with signed contribution d_c and overall total D, we also compute $100(D - d_c)/48$. This neutralizes its net contribution while holding the full denominator fixed. Across the inferred

Table 24: Paired differences in percentage points, with 95% cluster-block bootstrap intervals.

Contrast	ΔAcc.	ΔFQ1-GEM	ΔNTS	ΔDIS
InternVL3-8B : All–Text	-1.4 [-6.9,+4.2]	+0.0 [-5.9,+6.2]	+8.3 [+2.0,+17.0]	+12.5 [+0.0,+25.5]
InternVL3-8B : All–One	+1.7 [-3.4,+6.7]	+2.1 [+0.0,+7.0]	+2.1 [-6.8,+11.5]	-8.3 [-20.0,+2.2]
InternVL3-8B LoRA: All–Text	+2.4 [-3.5,+7.9]	+0.0 [+0.0,+0.0]	+8.3 [+2.0,+16.7]	+8.3 [+2.0,+16.7]
InternVL3-8B LoRA: All–One	+4.9 [+1.3,+8.8]	+0.0 [+0.0,+0.0]	+4.2 [-3.7,+12.8]	-2.1 [-12.8,+8.3]
InternVL3.5-14B : All–Text	-2.1 [-7.3,+3.1]	+0.0 [+0.0,+0.0]	+6.2 [+0.0,+13.3]	+10.4 [-2.1,+24.4]
InternVL3.5-14B : All–One	+3.5 [-1.8,+8.7]	+0.0 [+0.0,+0.0]	-2.1 [-9.3,+4.5]	+16.7 [+4.3,+29.5]
Kimi-VL-A3B : All–Text	+4.2 [-0.8,+9.3]	+0.0 [+0.0,+0.0]	+0.0 [+0.0,+0.0]	+4.2 [-7.0,+16.7]
Kimi-VL-A3B : All–One	-1.0 [-6.6,+4.7]	-2.1 [-6.7,+0.0]	-6.2 [-13.0,+0.0]	+8.3 [-3.7,+22.4]
LLaVA-OneVision-7B : All–Text	+0.0 [-5.3,+5.2]	+0.0 [+0.0,+0.0]	+0.0 [+0.0,+0.0]	-6.2 [-17.0,+4.1]
LLaVA-OneVision-7B : All–One	-2.4 [-6.7,+1.5]	+0.0 [+0.0,+0.0]	+0.0 [+0.0,+0.0]	-12.5 [-23.8,-3.6]
MiniCPM-V 4.5 : All–Text	+3.8 [-1.3,+9.7]	+0.0 [+0.0,+0.0]	+4.2 [+0.0,+10.2]	+14.6 [-2.0,+32.5]
MiniCPM-V 4.5 : All–One	+1.0 [-3.8,+6.4]	+0.0 [+0.0,+0.0]	-4.2 [-12.5,+3.9]	+4.2 [-6.8,+16.3]
Mistral Small 3.2 24B : All–Text	-3.8 [-7.1,-0.6]	+0.0 [+0.0,+0.0]	+2.1 [+0.0,+6.7]	+8.3 [+1.9,+17.1]
Mistral Small 3.2 24B : All–One	-3.8 [-9.1,+2.0]	+0.0 [+0.0,+0.0]	-4.2 [-12.8,+4.1]	-22.9 [-35.6,-10.0]
Phi-4-Multimodal : All–Text	-2.4 [-9.7,+4.9]	-2.1 [-6.7,+0.0]	+0.0 [+0.0,+0.0]	+16.7 [-4.2,+37.8]
Phi-4-Multimodal : All–One	-1.7 [-6.7,+3.4]	+0.0 [+0.0,+0.0]	-4.2 [-10.0,+0.0]	-2.1 [-12.5,+8.7]
Qwen2.5-VL-32B : All–Text	+2.4 [-4.7,+9.4]	+0.0 [+0.0,+0.0]	+6.2 [+0.0,+14.0]	+35.4 [+21.3,+51.0]
Qwen2.5-VL-32B : All–One	+6.6 [+1.3,+11.9]	+0.0 [+0.0,+0.0]	-2.1 [-9.1,+4.8]	+18.8 [+3.9,+34.8]
Qwen2.5-VL-7B : All–Text	-3.1 [-9.9,+3.9]	+0.0 [+0.0,+0.0]	+2.1 [+0.0,+6.8]	+43.8 [+29.2,+59.5]
Qwen2.5-VL-7B : All–One	-0.3 [-3.3,+3.3]	+0.0 [+0.0,+0.0]	+0.0 [+0.0,+0.0]	+6.2 [-4.2,+17.8]
Qwen2.5-VL-7B LoRA: All–Text	+3.1 [-2.0,+8.3]	-2.1 [-6.7,+0.0]	+12.5 [+4.2,+22.0]	+12.5 [+3.8,+23.4]
Qwen2.5-VL-7B LoRA: All–One	+4.2 [-0.3,+8.9]	+0.0 [+0.0,+0.0]	+12.5 [+4.3,+22.2]	+12.5 [+3.8,+23.4]
Qwen3-VL-8B : All–Text	+4.5 [-2.2,+11.4]	+0.0 [+0.0,+0.0]	+14.6 [+4.5,+26.2]	+33.3 [+18.2,+49.0]
Qwen3-VL-8B : All–One	+5.2 [+0.0,+10.7]	+0.0 [+0.0,+0.0]	+6.2 [-4.8,+18.9]	+29.2 [+17.0,+42.0]
Qwen2.5-VL-7B: LoRA–Base	+6.6 [-0.3,+12.8]	+0.0 [+0.0,+0.0]	+10.4 [+0.0,+20.9]	-31.2 [-51.2,-11.3]
InternVL3-8B: LoRA–Base	+3.1 [-0.7,+7.2]	-2.1 [-7.0,+0.0]	+0.0 [-8.5,+8.2]	-12.5 [-22.9,-3.8]

Unmarked models are base checkpoints. Positive values favor the first condition. Paired outcome differences are retained while exact-image asset-component fallback clusters are resampled as blocks.

Table 25: Post-hoc LoRA–Base sensitivity on 48 FQ2 groups. Changes and 95% intervals are in percentage points. Gain/loss gives paired counts. Intervals use 20,000 block-bootstrap draws over 29 exact-image, 14 inferred-scene or 13 union components, with each draw normalized by its sampled group count.

Family	Metric	Δ (pp)	Exact-image	Inferred-scene	Union	Gain/loss
Qwen2.5-VL-7B	NTS	$+10.4$	$[0.0, +20.9]$	$[-2.6, +19.0]$	$[-2.4, +19.2]$	6/1
Qwen2.5-VL-7B	DIS	-31.2	$[-51.1, -11.4]$	$[-56.7, -8.7]$	$[-57.1, -9.4]$	3/18
InternVL3-8B	NTS	0.0	$[-8.5, +8.3]$	$[-12.5, +8.2]$	$[-12.5, +8.1]$	2/2
InternVL3-8B	DIS	-12.5	$[-22.8, -2.5]$	$[-27.3, -2.5]$	$[-27.3, -2.4]$	0/6

clusters, Qwen's NTS range remains positive and both DIS ranges remain negative. The supplement includes per-group outcomes, cluster contributions and a separate verification implementation.

E Deletion and Image-Reference Audit

Offline reconstruction reproduces all 864 question–mode inputs and 130 prepared images without position labels. The evaluator adds neither stable original-image IDs nor missing-image placeholders. Both Circular input sets preserve source questions and image order across 1,152 rotations each. Reconstruction uses the saved implementation and prepared inputs rather than archived wire requests.

The runner uses current image order. Comparing the first explicit Chinese ordinal against the original and deleted lists identifies 36 removed referents and six retained at changed positions. All 36 removed referents are the original first image. Including groups 00368 and 00370 gives 38 ordinals pointing to another retained image. Groups 00360, 00361, 00364 and 00405 have out-of-range ordinals. Two retain the referent, while no explicit ordinal is detected in four groups.

For example, group 00392 still asks for the first image's camera direction after deleting that original image. Its request now begins with the original second image. The question text is unchanged, but its ordinal now points to a different image, coupling evidence removal with a reference change.

Sensitivity using the existing predictions. Table 26 links all 20 configurations to four exhaustive strata without replacing the 48-group main denominator. Two groups retain a stable explicit

Table 26: Deletion success across 48 groups stratified by the fate of the originally referenced image. M counts correct MSS answers and J jointly correct MSS/deletion answers. For a stratum of n groups, NTS is J/n and conditional deletion success is J/M when $M > 0$.

		Removed ($n = 36$)		Shifted ($n = 6$)		Stable ($n = 2$)		No ordinal ($n = 4$)	
Model	Setting	M	J	M	J	M	J	M	J
Claude Fable 5.1	0-shot	22	4	2	0	2	0	0	0
Claude Fable 5.1	1-shot	19	0	1	1	2	0	1	0
Claude Fable 5.1	2-shot	21	4	2	1	1	0	3	1
Claude Fable 5.1	3-shot	19	3	2	1	1	0	1	0
GPT-5.6	0-shot	20	5	3	1	2	1	1	0
GPT-5.6	1-shot	20	7	3	1	2	0	3	0
GPT-5.6	2-shot	20	8	5	3	0	0	2	0
GPT-5.6	3-shot	23	7	4	2	2	1	1	0
InternVL3-8B	Base	14	4	1	0	0	0	0	0
InternVL3-8B	LoRA	10	4	1	0	0	0	0	0
InternVL3.5-14B	Base	12	3	0	0	0	0	0	0
Kimi-VL-A3B	Base	20	0	0	0	0	0	2	0
LLaVA-OneVision-7B	Base	19	0	2	0	1	0	3	0
MiniCPM-V-4.5	Base	23	2	1	0	0	0	2	0
Mistral-Small-3.2	Base	3	0	0	0	0	0	1	1
Phi-4-Multimodal	Base	16	0	2	0	0	0	0	0
Qwen2.5-VL-32B	Base	15	1	1	0	1	1	2	1
Qwen2.5-VL-7B	Base	20	1	0	0	1	0	1	0
Qwen2.5-VL-7B	LoRA	7	5	0	0	1	1	0	0
Qwen3-VL-8B	Base	19	7	1	0	0	0	1	0

Table 27: GPT stable-ID prompting on the same twelve FQ2 triplets. Orig. and IDs count released-label matches. Changed counts answer differences between prompts in (a) and the left half of (b), and between blocks in the right half of (b).

(a) Counterbalanced prompt pairs

Reference status	n	Orig.	IDs	Changed	Input condition	n	Orig.	IDs	Changed
Retained, shifted	18	12	8	8	MSS	12	8	6	6
Stable ordinal	6	6	2	4	Deletion	12	7	4	6
No ordinal	12	6	5	6	Hard addition	12	9	5	6
Overall	36	24	15	18	Overall	36	24	15	18

(b) Two randomized execution blocks

Block	n	Orig.	IDs	Changed	Fixed prompt	n	Block 1	Block 2	Changed
1	36	17	17	12	Original	36	17	20	14
2	36	20	21	15	Stable IDs	36	17	21	15

ordinal, and four have no detected ordinal. For Qwen2.5-VL-7B, jointly correct deletion responses increase from one to five among the 36 removed-anchor groups and from zero to one among the two stable-ordinal groups. Both InternVL3-8B checkpoints obtain all four of their joint successes in the removed-anchor stratum. All four checkpoints fail jointly on the six retained-but-shifted groups.

Successes concentrate in removed-anchor groups rather than shifted-reference cases. The six stable-ordinal or no-ordinal groups are too few to establish the pattern independently.

Paired stable-ID prompting diagnostic. We select all twelve FQ2 groups outside the removed-anchor stratum, covering six retained-but-shifted groups, two stable-ordinal groups and four groups without an ordinal. Each group contributes MSS, deletion and hard-addition inputs. The paired diagnostic obtains GPT responses for both prompts, giving 36 items per arm and 72 valid responses. Images, their order, options and released labels are held fixed. The rewrite assigns stable original-image IDs, adds an explicit ID mapping and instructions, and replaces ordinal references with those IDs. Original wording is requested first on eighteen pairs and the rewrite first on eighteen. Execution is interleaved rather than simultaneous.

Table 27a reports released-label agreement and prediction changes. Twelve pairs match the la-

bel in both arms, twelve only under original wording, three only under the rewrite and nine in neither. The overall agreement difference is -25.00 points. Paired resampling of twelve groups gives a 95% percentile interval of $[-44.44, -5.56]$ from 10,000 draws with seed 20261010. Using the six image/scene union components gives $[-41.67, -3.03]$ with seed 20261110. The corresponding intervals for the 50.00% prediction-change rate are [30.56, 69.44]% and [25.00, 70.83]%.

Repeated-block control. The repeated-block control holds these 36 items and both prompt formulations fixed across two blocks, each containing all 72 item–prompt combinations. Within each block, requests are shuffled with seeds 20261012 and 20261013 and run with concurrency two. Block one finishes before block two starts. This randomizes interleaving without enforcing which prompt is requested first for each item. Images, their order, labels and decoding settings remain unchanged. Only transport or HTTP failures are retried, at most three attempts per request. All 144 responses are valid.

Across the blocks in Table 27b, original wording matches 37/72 labels and stable IDs match 38/72, a difference of 1.39 points. A paired bootstrap with 20,000 draws over the twelve groups and seed 20261011 retains all three conditions and both blocks within each group. Its 95% interval is $[-12.50, 16.67]$. Resampling the six image/scene union components instead gives $[-6.94, 16.67]$. These intervals condition on the two observed blocks. Across their 24 group–block observations, NTS falls from 8 to 4 and DIS from 9 to 8, while three-answer joint agreement remains 4. NTS has two paired gains and six losses despite near-equal item agreement.

Original prompts change 14/36 answers between blocks, and stable-ID prompts change 15/36. The initial 25-point decline therefore does not recur in the repeated comparison. Response instability can coexist with nearly equal aggregate agreement, so a single paired run does not isolate a wording effect. The intervention combines IDs, mapping instructions and visible numbering gaps rather than changing ID binding alone.

Targeted machine-assisted visual check. Given exact prompts and ordered images from six priority groups, two isolated contexts of one assistant save answers before seeing gold labels. Both match labels on the same 11 of 18 originals and abstain on seven owing to entity or view ambiguity. Audit abstention means inability to adjudicate, not an insufficient-information answer. For nine additional single-view deletions, both select insufficiency on six and abstain on three. Four original and five additional items have out-of-range references that prevent judging necessity with unchanged bindings. The saved responses and visual rationales localize entity and reference ambiguity in this six-group model-assisted check.

E.1 Original Inputs and Image-Order Traces

Figure 7 reproduces two Chinese FQ2 triplets with ordered images and English translations. Source IDs identify images for comparison, while the evaluated requests use their current order. The complete input audit covers all 48 test triplets and their 144 released conditions. Removing one image preserves the order of the others but can change which image an ordinal refers to. Group 00392 illustrates removal of the original anchor, and group 00368 illustrates a retained anchor whose position changes. Both retain the evaluated prompts and labels.

F Offline Diagnostic Supplements

F.1 Sensitivity to Scene/Template Overlap

The inferred scene/template ledger and fixed split define overlap with train, dev alone, or neither. These categories contain 81, zero and 15 test groups, respectively. Rates are omitted for the empty dev-only stratum.

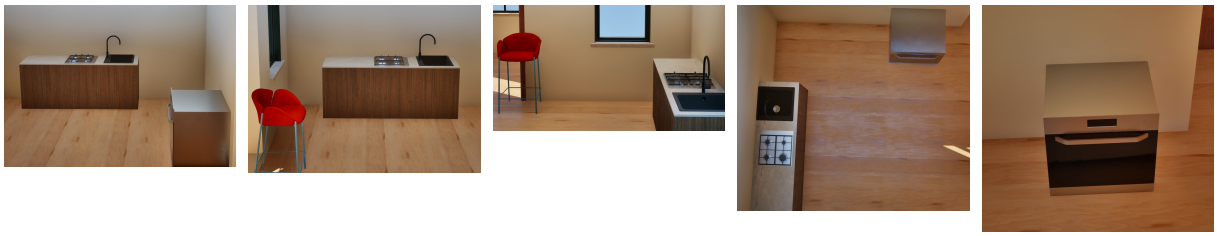
The overlap stratum contains 41 FQ1 and 40 FQ2 groups across ten clusters. The four non-overlapping clusters contain seven FQ1 and eight FQ2 groups. Their 13 synthetic and two real groups contrast with 41 synthetic and 40 real overlap groups. These strata measure descriptive sensitivity rather than a causal scene-exposure effect.

Table 28 preserves all 20 primary configurations. Every primary configuration has zero GEM on the seven non-overlapping FQ1 groups. Qwen's primary LoRA reduces DIS in both strata, from 17/40 to 4/40 in overlapping groups and from 4/8 to 2/8 elsewhere. InternVL's NTS changes from 3/40 to 2/40 and from 1/8 to 2/8, respectively. The unchanged overall NTS therefore combines opposing subgroup changes. Table 29 separately reports matched bases and three training seeds.

Paired group outcomes further distinguish gains from losses. Qwen's non-overlapping DIS decline comprises two gains and four losses. Across strata,

(a) Original anchor removed - group 00392

S01 S02 S03 S04 S05



以图一镜头朝向为正前方，面对前方时右手侧为右方，烤箱位于椅子的哪一侧？

Take the camera's viewing direction in Image 1 as forward, and the right-hand side when facing forward as right. On which side of the chair is the oven?

A. 前方 / Front B. 右方 / Right C. 后方 / Behind D. 信息不足 / Insufficient information

Input	Actual image sequence	Ordinal now refers to	Released label
MSS	S01, S02, S03, S04	S01	B
Deletion	S02, S03, S04	S02	D
Addition	S01, S02, S03, S04, S05	S01	B

The released deletion removes S01. The first image delivered to the model is then S02, while the unchanged question still refers to Image 1.

(b) Original anchor retained at a new position - group 00368

S01 S02 S03 S04



以图二镜头朝向为正前方，面对前方时右手侧为右方，盆栽位于桌子的哪一侧？

Take the camera's viewing direction in Image 2 as forward, and the right-hand side when facing forward as right. On which side of the table is the potted plant?

A. 右后方 / Back-right B. 右前方 / Front-right C. 信息不足 / Insufficient information D. 左后方 / Back-left

Input	Actual image sequence	Ordinal now refers to	Released label
MSS	S01, S02, S03	S02	A
Deletion	S02, S03	S03	C
Addition	S01, S02, S03, S04	S02	A

The released deletion removes S01 and retains the original anchor S02 in the first position. The second image delivered to the model is now S03, while the question still refers to Image 2.

Figure 7: Original Chinese questions, options and image order in two FQ2 triplets. English text translates the questions for reading. S identifiers track source images, while the original inputs use unlabeled current-order images. Labels are those of the released dataset.

numerators, denominators, gains and losses sum exactly to their full-test counterparts. With eight FQ2 groups, each success changes the non-overlapping rate by 12.5 points.

F.2 Semantic Changes Across FQ1 Pairs

Each of the 48 FQ1 triplets contributes camera–object, camera–world and object–world pairs. Predicted letters are mapped to direction meanings. The 144 pairs per configuration are partitioned by gold-meaning changes and predicted changes, un-

Table 28: Scene/template overlap sensitivity of the primary configurations.

	Cluster shared with train				Cluster absent from train and dev			
Model / setting	Acc.	GEM	NTS	DIS	Acc.	GEM	NTS	DIS
Qwen2.5-VL-7B	28.40	0	1	17	31.11	0	0	4
InternVL3-8B	27.57	1	3	6	40.00	0	1	4
Qwen3-VL-8B	33.33	0	5	15	37.78	0	2	4
LLaVA-OneVision-7B	29.22	0	0	16	28.89	0	0	2
MiniCPM-V 4.5	32.92	0	2	19	33.33	0	0	4
Mistral Small 3.2 24B	25.10	0	1	4	28.89	0	0	0
Qwen2.5-VL-32B	33.74	0	3	14	31.11	0	0	3
InternVL3.5-14B	28.40	0	3	11	26.67	0	0	1
Phi-4-Multimodal	26.34	0	0	14	26.67	0	0	3
Kimi-VL-A3B	27.16	0	0	14	40.00	0	0	5
Qwen2.5-VL-7B / LoRA	34.57	0	5	4	40.00	0	1	2
InternVL3-8B / LoRA	30.45	0	2	3	44.44	0	2	1
GPT-5.6 / 0-shot	41.56	2	6	15	35.56	0	1	2
1-shot	42.39	3	6	16	53.33	0	2	4
2-shot	46.91	5	6	15	55.56	0	5	6
3-shot	51.03	3	8	19	44.44	0	2	4
Claude Fable 5.1 / 0-shot	38.68	0	3	17	42.22	0	1	3
1-shot	40.74	3	1	15	37.78	0	0	3
2-shot	40.33	3	5	16	46.67	0	1	2
3-shot	42.39	1	2	16	37.78	0	2	3

Acc. is a percentage over 243 or 45 questions. GEM is a count out of 41 or 7 FQ1 groups. NTS and DIS are counts out of 40 or 8 FQ2 groups. The partition uses an inferred scene/template ledger. No test cluster overlaps only dev. Invalid answers remain errors. Unmarked open models are bases.

changed meanings or invalid responses. Joint correctness is counted within each cell.

All FQ1 test gold pairs require a semantic change, and all predictions are valid. False-change rates on unchanged gold therefore cannot be estimated. The 2,880 configuration-pair records reuse the same 48 groups.

Table 30 separates answer changes from successful switching. GPT two- and three-shot each change 94 pairs, with 26 and 27 jointly correct, yet fully correct triplets fall from five to three. Improved pairwise correctness need not extend to all three frames. Qwen3 changes only 29 pairs, with one jointly correct.

F.3 Train-Only Text Shortcut Baseline

We fit a single shallow classifier with fixed hyperparameters to the visible question followed by its A–D option texts. Character TF-IDF uses 2–5-grams, minimum document frequency two, a 50,000-feature cap, sublinear term frequency and L2 normalization, without lowercasing.

Logistic regression uses $C = 1$, L-BFGS, no class weighting, at most 5,000 iterations and seed 20261011. Vocabulary, inverse document frequencies and classifier weights are learned from the 576 training items. Features exclude identifiers, condition tags, image counts and paths.

The fitted vocabulary contains 3,623 features, and optimization converges in 31 iterations without warnings. The 96 dev items provide a descriptive check without tuning. One evaluation on the 288 test items gives 69 correct answers. Table 31 reports macro classification metrics and joint counts from the same saved predictions. Implementation uses scikit-learn 1.7.2 on CPU.

The fixed text classifier performs near the four-choice accuracy reference and obtains no fully correct FQ1 groups or MSS–deletion pairs. It nevertheless solves five MSS–addition pairs, including all five groups on which its MSS answer is correct. Addition agreement can thus persist even for a weak text-only classifier.

Table 29: Scene/template overlap sensitivity of the separate three-seed cohort.

	Cluster shared with train				Cluster absent from train and dev			
Model / setting	Acc.	GEM	NTS	DIS	Acc.	GEM	NTS	DIS
InternVL3-8B / Base control	28.40	1	3	6	40.00	0	1	4
InternVL3-8B / LoRA 20260722	30.86	0	3	6	44.44	0	2	1
InternVL3-8B / LoRA 20260723	28.81	0	2	4	42.22	0	2	1
InternVL3-8B / LoRA 20260724	30.86	0	3	4	46.67	1	2	2
Qwen2.5-VL-7B / Base control	28.81	0	1	18	31.11	0	0	4
Qwen2.5-VL-7B / LoRA 20260722	34.98	0	3	4	40.00	0	1	2
Qwen2.5-VL-7B / LoRA 20260723	33.74	0	3	4	40.00	0	1	2
Qwen2.5-VL-7B / LoRA 20260724	37.45	0	6	4	37.78	0	1	2

Acc. is a percentage over 243 or 45 questions. GEM is a count out of 41 or 7 FQ1 groups. NTS and DIS are counts out of 40 or 8 FQ2 groups. The partition uses an inferred scene/template ledger. No test cluster overlaps only dev. Seed numbers identify LoRA adapters, with a separate matched base for each family.

Table 30: FQ1 semantic changes across frame pairs.

Model / setting	Same	Change	Both	GEM
Qwen2.5-VL-7B	114	30	4	0
InternVL3-8B	94	50	4	1
Qwen3-VL-8B	115	29	1	0
LLaVA-OneVision-7B	116	28	4	0
MiniCPM-V 4.5	99	45	3	0
Mistral Small 3.2 24B	112	32	2	0
Qwen2.5-VL-32B	103	41	1	0
InternVL3.5-14B	111	33	1	0
Phi-4-Multimodal	120	24	1	0
Kimi-VL-A3B	108	36	4	0
Qwen2.5-VL-7B / LoRA	103	41	6	0
InternVL3-8B / LoRA	113	31	6	0
GPT-5.6 / 0-shot	54	90	18	2
1-shot	54	90	21	3
2-shot	50	94	26	5
3-shot	50	94	27	3
Claude Fable 5.1 / 0-shot	92	52	7	0
1-shot	68	76	23	3
2-shot	59	85	17	3
3-shot	65	79	15	1

Same and Change count predicted direction relations out of 144 frame pairs. Both counts pairs with two correct answers, a subset of Change. GEM counts fully correct triplets out of 48. All evaluated test gold pairs require a direction change and all responses are valid. Unmarked open models are bases.

Table 31: Train-fitted question-and-option text baseline.

Metric	Dev (96 questions)	Test (288 questions)
P	25.29	23.99
R	24.76	23.97
F1	24.63	23.97
Acc.	25.00	23.96
GEM	0/16	0/48
NTS	0/16	0/48
DIS	4/16	5/48

P, R and F1 are macro-averaged percentages over A–D. GEM, NTS and DIS are joint-success counts. Character TF-IDF and logistic regression are fitted on 576 training questions with fixed settings. Dev does not guide model selection.